

RENAISSANCE UNIVERSITY, INDORE

SCHOOL OF COMPUTER SCIENCE

Subject name: Website Designing; Subject Code: BCA-405

UNIT -1

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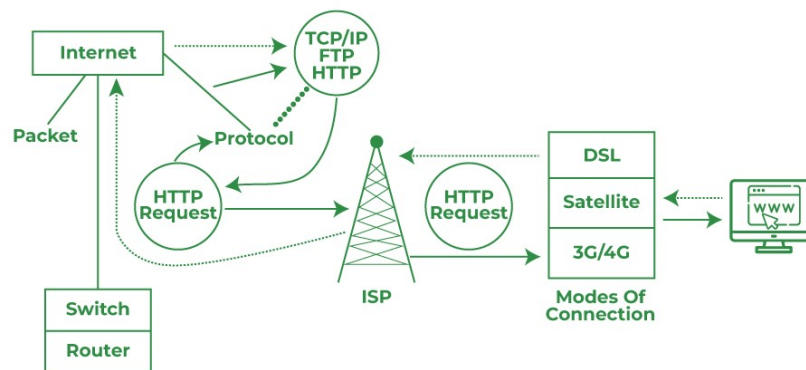
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1. Internet

The Internet is a global network that uses TCP / IP protocols to connect millions of computers worldwide. It can include different networks, such as private, public, organizational, academic, government, etc. It allows users to communicate with each other and share massive amounts of data in various formats. Internet is available in both modes, wired and wireless. In wired mode, the data travels through fiber optic cables, whereas in wireless mode, the data travels through radio waves.

1.1 Working of Internet

From opening a web browser to visiting a website, it all happens with specific methods that we're going to check in these **5 easy steps**.



1. Firstly, you'll be required to connect your system or PC with any router or modem to establish a connection. This connection is the base of the internet connection.
2. When you open the browser and start typing something like "www.google.com", your system will push a query command to your ISP (Internet Service Provider) that is connected with other servers that store and process data.
3. Now, the web browser will start indexing the URL that you've entered and will fetch the details in numeric format (in their language to identify the address (unique) that you're trying to reach.
4. Next, now your browser will start sending the HTTP request where you're trying to reach and send a copy of the website on the user's system.
5. Once all the data (of small packets) is received at the user's end (PC/Laptop), the browser will start arranging all those small packets and later will form a collective file (here, the

browser will gather all the small packets and rearrange them just like a puzzle) and then you'll be able to see the contents of that website.

1.2 Advantages of Internet

- Global connections through virtual communications,
- E-commerce,
- Online Education,
- Abundance of information

1.3 Disadvantages of Internet

- Loss of personal information,
- Spread of fake news,
- Physical and mental health issues,
- Internet addiction and time wastage

1.4 Applications of Internet

- a) File sharing,
- b) Downloading media files and software,
- c) Sending and receiving emails,
- d) Browsing any information ,
- e) Using social media platforms, forums, and communities ,
- f) E-Commerce, bill payments, online purchasing of food and medicines ,
- g) Voice Calls, Video Conferencing, and Chatting with friends, family members, and colleagues

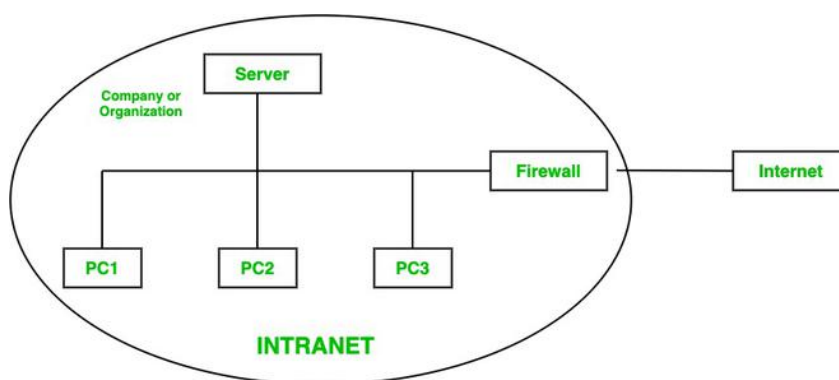
1.5 History of Internet

- **1969:** Launch of **ARPANET**, the first operational packet-switching network.
- **1971:** The first **email** is sent by **Ray Tomlinson**.
- **1983:** Adoption of **TCP/IP** as the standard protocol for ARPANET.
- **1989-1990:** **Tim Berners-Lee** invents the **World Wide Web**.
- **1993:** The first graphical web browser, **Mosaic**, is released.
- **1998:** **Google** is founded, revolutionizing search engines.
- **2007:** The **smart phone** introduces widespread mobile Internet access.
- **2010s:** Rise of **cloud computing**, **social media**, and **streaming services**.

2. Intranet

An Intranet is a part of the Internet and is owned and used privately by an organization. It is mainly used to connect all the computers and establish a private network of an organization to provide employees the ability to collaborate on projects, manage or update information, share calendars, and to-do list, etc. Organizations prefer using Intranet to keep their data inaccessible from outsiders, making their suspicious data and project information secure. Intranet includes a firewall to prevent unauthorized users from accessing the network.

2.1 Working of Intranet



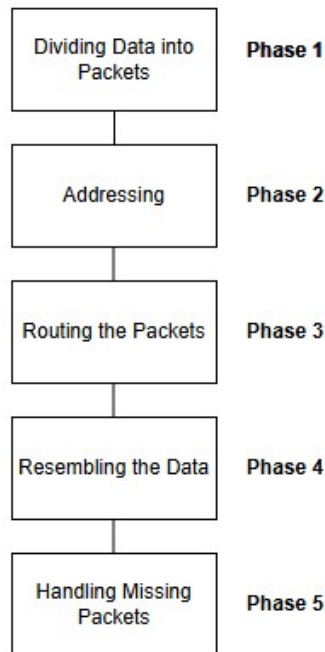
3. Difference between Internet and Intranet

Feature	Internet	Intranet
Definition	A global network connecting millions of computers worldwide.	A private network used within an organization.
Access	Open to the public, accessible from anywhere.	Restricted to specific users (employees, members) within an organization.
Users	Anyone with an internet connection.	Only authorized personnel of an organization.
Security	More vulnerable to cyber threats; requires strong security measures.	More secure as it operates within a closed environment.
Purpose	Used for communication, research, entertainment, business, and more.	Used for internal communication, sharing documents, and collaboration.
Examples	Websites, social media, online shopping, email.	Company portals, internal HR systems, private messaging platforms.

4. Internet Protocol

It is a set of rules that allows devices to communicate with each other over the Internet. It is like the address system used for sending data. Every device connected to the internet has a unique **IP address** that helps data know where to go and where it is coming from.

4.1 Working of Internet Protocol

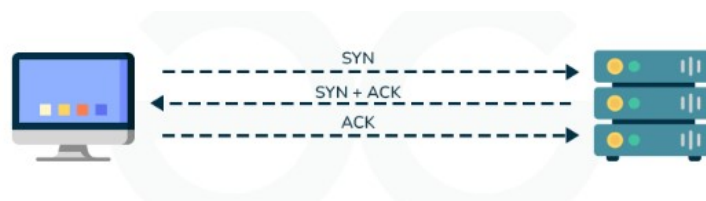


5. Types of Internet Protocol

Internet Protocols are of different types having different uses. These are mentioned below:

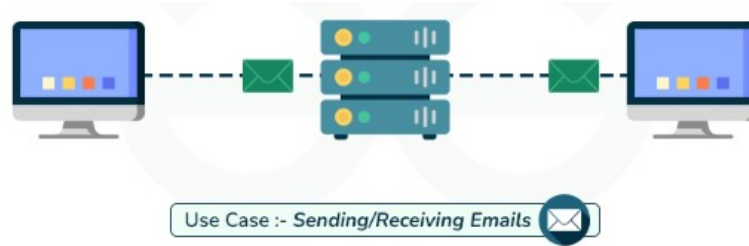
5.1 TCP/IP (Transmission Control Protocol/ Internet Protocol)

In TCP/IP, the IP protocol ensures that each computer that is connected to the Internet is having a specific serial number called the IP address. TCP specifies how data is exchanged over the internet and how it should be broken into IP packets. It also makes sure that the packets have information about the source of the message data, the destination of the message data, the sequence in which the message data should be re-assembled, and checks if the message has been sent correctly to the specific destination. The TCP is also known as a connection-oriented protocol. (Here ACK = Acknowledgement, & SYN = Synchronization)



5.2 SMTP (Simple Mail Transfer Protocol)

SMTP protocol is important for sending and distributing outgoing emails. This protocol uses the header of the mail to get the email id of the receiver and enters the mail into the queue of outgoing mail. And as soon as it delivers the mail to the receiving email id, it removes the email from the outgoing list.



5.3 PPP (Point-to-Point Protocol)

PPP is a communication protocol that is used to create a direct connection between two communicating devices. This protocol defines the rules using which two devices will authenticate with each other and exchange information with each other.

5.4 FTP (File Transfer Protocol)

This protocol is used for transferring files from one system to the other. This works on a client-server model. When a machine requests for file transfer from another machine, the FTO sets up a connection between the two and authenticates each other using their ID and Password. And, the desired file transfer takes place between the machines.



5.5 SFTP (Secure File Transfer Protocol)

SFTP encrypts both commands and data while in transmission.

5.6 HTTP (Hyper Text Transfer Protocol)

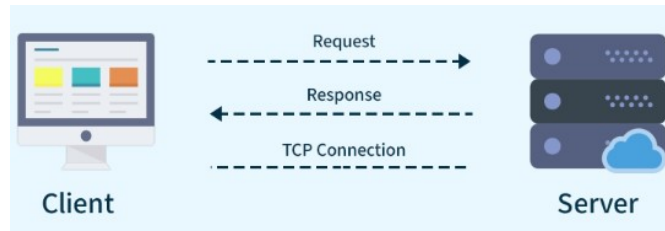
HTTP protocol is used to transfer hypertexts over the internet and it is defined by the www (World Wide Web) for information transfer. This protocol defines how the information needs to be formatted and transmitted. And, it also defines the various actions the web browsers

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should take in response to the calls made to access a particular web page. Whenever a user opens their web browser, the user will indirectly use HTTP as this is the protocol that is being used to share text, images, and other multimedia files on the World Wide Web.



5.7 HTTPS (Hypertext Transfer Protocol Secure)

HTTPS is an extension of the Hypertext Transfer Protocol (HTTP). It is used for secure communication over a computer network with the SSL/TLS (Secure Sockets Layer and Transport Layer Security) protocol for encryption and authentication.

5.8 TELNET

TELNET is a type of protocol that enables one computer to connect to the local computer. It is used as a standard **TCP/IP protocol** for virtual terminal service which is provided by **ISO**. The computer which starts the connection is known as the **local computer**. The computer which is being connected to i.e. which accepts the connection known as the **remote computer**. During telnet operation, whatever is being performed on the remote computer will be displayed by the local computer. Telnet operates on a client/server principle.

5.8.1 Logging in TELNET

The logging process can be further categorized into two parts:

- Local Login
- Remote Login

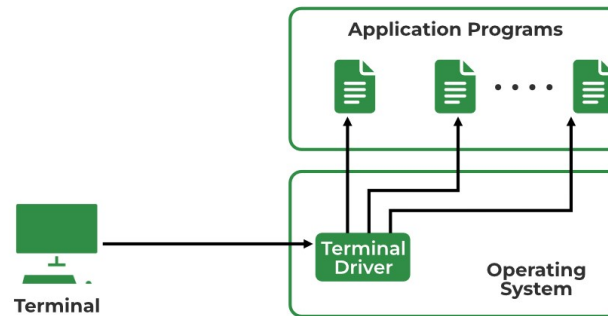
a) Local Login

Whenever a user logs into its local system, it is known as local login.

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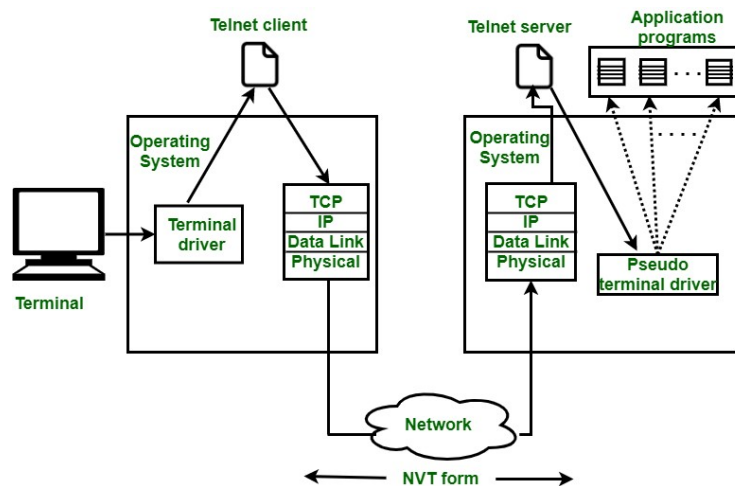


The Procedure of Local Login

- Keystrokes are accepted by the terminal driver when the user types at the terminal.
- Terminal Driver passes these characters to OS.
- Now, OS validates the combination of characters and opens the required application.

b) Remote Login

Remote Login is a process in which users can log in to a remote site i.e. computer and use services that are available on the remote computer. With the help of remote login, a user is able to understand the result of transferring the result of processing from the remote computer to the local computer.

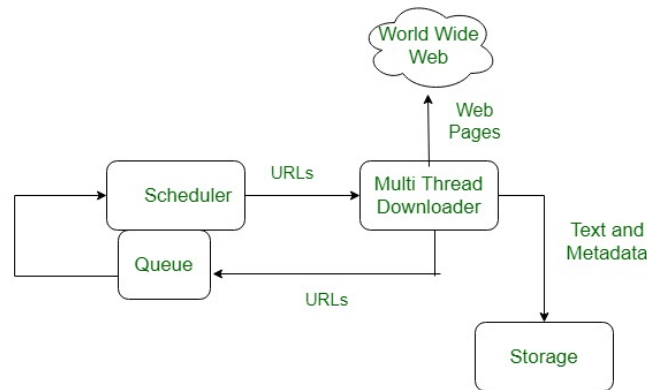


6. WWW (World Wide Web)

WWW stands for World Wide Web and is commonly known as the Web. The WWW was started by CERN in 1989. WWW is defined as the collection of different websites around the world, containing different information shared via local servers (or computers).

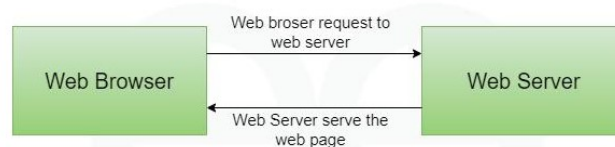
6.1 System Architecture OF WWW

From the user's point of view, the web consists of a vast, worldwide connection of documents or web pages. Each page may contain links to other pages anywhere in the world. The pages can be retrieved and viewed by using browsers of which internet explorer, Netscape Navigator, Google Chrome, etc are the popular ones. The browser fetches the page requested interprets the text and formatting commands on it, and displays the page, properly formatted, on the screen. The basic model of how the web works are shown in the figure below. Here the browser is displaying a web page on the client machine. When the user clicks on a line of text that is linked to a page on the abd.com server, the browser follows the hyperlink by sending a message to the abd.com server asking it for the page.



Here the browser displays a web page on the client machine when the user clicks on a line of text that is linked to a page on abd.com, the browser follows the hyperlink by sending a message to the abd.com server asking for the page.

6.2 Working of WWW



7. DNS (Domain Name System)

The Domain Name System is a hierarchical and distributed name service that provides a naming system for computers, services, and other resources on the Internet or other Internet Protocol networks. It is the phonebook of the Internet, connecting web browsers with websites.



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8. URL (Uniform Resource Locator)

Uniform Resource Locator, is an address that specifies the location of a resource on the internet. URLs are used to find web pages, images, videos, and more.

*****End of Unit*****

UNIT RELATED QUESTION

Q.1 Define: i) Internet, ii) Intranet, iii) URL, iv) DNS, v) HTTP, vi) HTTPS, vii) FTP, viii) SMTP, ix) PPP, x) TCP/IP.

Q.2 Mention the history of internet.

Q.3 What is internet? Mention the advantages, disadvantages & applications of it.

Q.4 Explain the working of internet with diagram in detail.

Q.5 What is intranet? Draw its working structure.

Q.6 Write the difference between internet and intranet.

Q.7 Write the types of internet protocol and explain any one of them.

Q.8 Explain the system architecture & working of WWW.

Q.9 Write a short note on remote logging in TELNET with diagram.

Q.10 Write a short note on local logging in TELNET with diagram.

Q.11 Write a short note on Internet Protocol & its working with diagram.

UNIT -2 (HTML)

- **Introduction to HTML.**
- **Basic structure of an HTML document. Creating an HTML document.**
- **HTML elements and tags,**
- **Formatting with HTML tags.**
- **Working with text, lists, tables and multimedia.**
- **Working with Forms and controls**

What is HTML?

HTML stands for Hypertext Markup Language and it is used to create WebPages. It uses HTML tags and attributes to describe the structure and formatting of a web page.

HTML consists of various elements that are responsible for telling search engines how to display page content. For example, headings, lists, images, links, and more.

HTML Example

```
<!DOCTYPE html>
<html>
  <head>
    <title>First HTML Code</title>
  </head>
  <body>
    <h2>Welcome To my first website </h2>
    <p>Hello </p>
  </body>
</html>
```

Features of HTML

- It is easy to learn and easy to use.
- It is platform-independent.
- Images, videos, and audio can be added to a web page.
- Hypertext can be added to the text.
- It is a markup language.

HTML Elements and Tags

HTML uses predefined tags and elements that tell the browser how to display the content.

HTML elements include an opening tag, some content, and a closing tag.

Remember to include closing tags. If omitted, the browser applies the effect of the opening tag until the end of the page.

Opening Tag	Content	Closing
↓	↓	↓
<code><p></code>	Hello World!	<code></p></code>

Breakdown:

- `<p>` → Opening tag (starts the paragraph)
- Hello World! → Content inside the tag
- `</p>` → Closing tag (ends the paragraph)

HTML Page Structure

The basic structure of an HTML page is shown below. It contains the essential building-block elements (i.e. doctype declaration, HTML, head, title, and body elements) upon which all web pages are created.

HTML Page Structure

```

<!DOCTYPE html>    ← Tells version of HTML
<html>             ← HTML Root Element

<head>             ← Used to contain page HTML metadata
  <title>Page Title</title> ← Title of HTML page
</head>

<body>             ← Hold content of HTML
  <h2>Heading Content</h2> ← HTML heading tag
  <p>Paragraph Content</p> ← HTML paragraph tag
</body>

</html>
  
```

HTML Structure & Tags

S.No	Tag	Description	Example
1	<!-- Comment -->	Used to insert comments in HTML.	<!-- This is a comment -->
2	<!DOCTYPE html>	Declares the document type (HTML5). Not a tag.	<!DOCTYPE html>
3	<html>	Root of the HTML document.	<html> ... </html>
4	<head>	Contains meta info, links, and scripts. Not visible on the page.	<head> ... </head>

S.No	Tag	Description	Example
5	<style>	Adds CSS styles within HTML.	<style> body {color: red;} </style>
6	<title>	Title shown on browser tab.	<title>My Webpage</title>
7	<base>	Specifies base URL for all relative URLs.	<base href="https://example.com/">
8	<noscript>	Content shown if JavaScript is disabled.	<noscript>JS Disabled</noscript>
9	<script>	Embeds JavaScript.	<script> alert("Hi"); </script>
10	<meta>	Defines metadata like character set, description, etc. (self-closing)	<meta charset="UTF-8">
11	<link>	Links external resources like CSS. (self-closing)	<link rel="stylesheet" href="styles.css">
12	<body>	Contains all visible content on the webpage.	<body> ... </body>

Common HTML Elements

S.No	Tag	Description	Example
1	<h1> to <h6>	Headings from highest (<h1>) to lowest (<h6>) level.	Main Heading
2	<p>	Defines a paragraph.	This is a paragraph.
3	<a>	Creates a hyperlink.	Visit Site
4		Embeds an image. (self-closing)	
5	<div>	Block-level container for grouping content.	<div class="section">Content here</div>
6		Inline container for text or style.	Blue text
7		Creates an unordered (bulleted)	- Item 1 - Item 2

S.No	Tag	Description	Example
		list.	
8		Creates an ordered (numbered) list.	- First - Second
9		Represents a list item.	List item
10	<table>	Creates a table.	<table> ... </table>
11	<tr>	Defines a row in a table.	<tr> ... </tr>
12	<td>	Defines a cell in a table row.	<td>Cell content</td>
13	<form>	Defines an HTML form for user input.	<form action="submit.php"> ... </form>
14	<input>	Creates an input field. (<i>self-closing</i>)	<input type="text" name="username">

Tags v/s Elements v/s Attributes difference

Feature	HTML Tag	HTML Element	HTML Attribute
Definition	A keyword enclosed in angle brackets (< >) that defines an element's beginning or end.	A complete structure that includes an opening tag, content, and a closing tag (or a self-closing tag).	Provides additional information or settings for an HTML element.
Components	Opening tag (<tagname>) and closing tag (</tagname>) or a self-closing tag.	Opening tag, content, and closing tag (or just a self-closing tag).	Name-value pairs written inside an opening tag (name="value").
Usage	Defines the start and end of an element.	Contains both tags and content.	Modifies the behavior or appearance of an element.
Example 1	and	Hello, World!	href="https://example.com" in Click Here
Example 2	 (self-closing tag)		src="image.jpg" and alt="Image" in

HTML Formatting

HTML Formatting is a process of formatting text for better look and feel. HTML provides us ability to format text without using CSS.

There are many formatting tags in HTML. These tags are used to make text bold, italicized, or underlined. There are almost 14 options available that how text appears in HTML and XHTML.

In HTML the formatting tags are divided into two categories:

- **Physical tag:** These tags are used to provide the visual appearance to the text.
- **Logical tag:** These tags are used to add some logical or semantic value to the text.

HTML Formatting Tags Table

HTML Formatting Tags

S.No	Tag	Definition (Simple)	Exemple
1		Makes the text bold.	This is bold text
2		Shows important text (usually bold)	This is strong text
3	<i>	Makes the text italic.	<i>This is italic text</i>
4		Emphasizes text (usually italic)	This is emphasized text
5	<mark>	Highlighteg text with a background	<mark>Highlighted text</mark>
6	<u>	Underlines the text.	<u>This is underlined</u>
7	<tt>	Shows text in teletype/monospace font (Not in HTML5)	<tt>Teletype text</tt>
8	<strike>	Draws a line through he text	<strike>Strikethrough text</strike>
9	<sup>	Displays text slightly above (super-	x²
10		Shows deleted text with underline	H₂O
11	<ins>	Marks added text with underline	Deleted text
12	<big>	Increases the font size slightly	<big>Bigger text</big>
14	<small>	Decreases the font size slightly	Smaller text

Example for HTML Formatting Tags

```
<html>
<head>
  <title>HTML Formatting Tags Example</title>
</head>
<body>

  <h1>HTML Formatting Tags Demo</h1>

  <p> <b>1. Bold Text</b> </p>
  <p> <strong>2. Strong (Important) Text</strong> </p>
  <p> <i>3. Italic Text</i></p>
  <p> <em>4. Emphasized Text </em> </p>
  <p> <mark>5. Highlighted Text </mark> </p>
  <p> <u>6. Underlined Text</u> </p>
  <p> <tt>7. Teletype Text (Monospace)</tt> </p>
  <p> <strike>8. Strikethrough Text</strike> </p>
  <p>9. Superscript: x<sup>2</sup> </p>
  <p>10. Subscript: H<sub>2</sub>O</p>
  <p> <del>11. Deleted Text</del> </p>
  <p> <ins>12. Inserted Text</ins> </p>
  <p> <big>13. Bigger Text</big> </p>
  <p> <small>14. Smaller Text</small> </p>

</body>
</html>
```

HTML Lists

An HTML list is a record of related information used to display the data or any information on web pages in the ordered or unordered form.

```
<!DOCTYPE html>
<html>
<head>
  <title>BCA IV SEM SUBJECTS</title>
</head>
<body>

  <h2>BCA IV SEM SUBJECTS</h2>

  <p> <strong>Unordered List:</strong> </p>
  <ul>
    <li>Java</li>
    <li>Website Designing</li>
    <li>AI</li>
  </ul>

  <p> <strong> Ordered List:</strong> </p>
  <ol>
    <li>Java</li>
    <li>Website Designing</li>
    <li>AI</li>
  </ol>

</body>
</html>
```

HTML List Tags

Here is the list of all **lists tags HTML**:

Tag	Description
<code></code>	Defines an unordered list
<code></code>	Defines an ordered list
<code></code>	Defines a list item
<code><dl></code>	Defines a description list
<code><dt></code>	Defines a term in a description list
<code><dd></code>	Details the term in a description list

The HTML Unordered List or Bulleted List

The [unordered list](#) items are marked with bullets. It is also known as bulleted lists. An unordered list starts with the ** tag**. Each list item starts with the ** tag**.

Syntax:

```
<ul> list of items </ul>
```

Attribute: This tag contains two attributes which are listed below:

- **[compact](#):** It will render the list smaller.
- **[type](#):** It specifies which kind of marker is used in the list.

Example:

```
<!DOCTYPE html>
<html>
<head>
  <title>BCA IV SEM SUBJECTS</title>
</head>
<body>

  <h2>BCA IV SEM SUBJECTS</h2>
```

```
<p><strong>Unordered List:</strong></p>
<ul>
  <li>Java</li>
  <li>Website Designing</li>
  <li>AI</li>
</ul>

<p><strong>Ordered List:</strong></p>
<ol>
  <li>Java</li>
  <li>Website Designing</li>
  <li>AI</li>
</ol>

</body>
</html>
```

Output:

BCA IV SEM SUBJECTS

Unordered List:

- Java
- Website Designing
- AI

Ordered List:

1. Java
2. Website Designing
3. AI

HTML Description List

A [description list](#) is a list of terms, with a description of each term. The <dl> tag defines the description list, the <dt> tag defines the term name, and the <dd> tag describes each term.

Syntax:

```
<dl> Contents... </dl>
details.
```



```
<!DOCTYPE html>
<html>
<head>
  <title>Grocery List</title>
</head>
<body>

  <h2>Grocery List with Quantities</h2>

  <dl>
    <dt>Apples</dt>
    <dd>2 kilograms</dd>

    <dt>Milk</dt>
    <dd>1 liter</dd>

    <dt>Bread</dt>
    <dd>2 loaves</dd>

    <dt>Rice</dt>
    <dd>5 kilograms</dd>
  </dl>

</body>
</html>
```

Output:

Grocery List with Quantities

Apples
2 kilograms
Milk
1 liter
Bread
2 loaves
Rice
5 kilograms

HTML Tables

An HTML Table is an arrangement of data in rows and columns in tabular format. Tables are useful for various tasks, such as presenting text information and numerical data. A table is a useful tool for quickly and easily finding connections between different types of data. Tables are also used to create databases.

HTML Table Tags with Description and Example

S.No.	Tag	Description	Example
1	<table>	Creates a table	<table> ... </table>
2	<tr>	Defines a table row	<tr> ... </tr>
3	<th>	Defines a header cell (bold & centered by default)	<th>Subject</th>
4	<td>	Defines a standard cell in a table	<td>Math</td>
5	<caption>	Adds a title/caption for the table	<caption>Student Marks</caption>
6	<thead>	Groups the header content in a table	<thead> <tr><th>...</th></tr> </thead>
7	<tbody>	Groups the main body content in a table	<tbody> <tr><td>...</td></tr> </tbody>
8	<tfoot>	Groups the footer content in a table	<tfoot> <tr><td>...</td></tr> </tfoot>
9	colspan	Makes a cell span multiple columns	<td colspan="2">Merged Cell</td>
10	rowspan	Makes a cell span multiple rows	<td rowspan="2">Merged Cell</td>

Simple Example of a Table in HTML:

```
<!DOCTYPE html>
<html>

<body>
  <table>
    <tr>
      <th>Firstname</th>
      <th>Lastname</th>
      <th>Age</th>
    </tr>
    <tr>
      <td>Priya</td>
      <td>Sharma</td>
      <td>24</td>
    </tr>
    <tr>
      <td>Arun</td>
      <td>Singh</td>
      <td>32</td>
    </tr>
    <tr>
      <td>Sam</td>
      <td>Watson</td>
      <td>41</td>
    </tr>
  </table>
</body>

</html>
```

Firstname	Lastname	Age
Priya	Sharma	24
Arun	Singh	32
Sam	Watson	41

Adding a border to an HTML Table

A border is set using the CSS border property. If you do not specify a border for the table, it will be displayed without borders.

Syntax

```
table, th, td {  
    border: 1px solid black;  
}
```

```
<!-- index.html -->  
<!DOCTYPE html>  
<html>  
  
<head>  
    <style>  
  
        table,  
        th,  
        td {  
            border: 1px solid black;  
        }  
    </style>  
</head>  
  
<body>  
    <h1 style="text-align: center;"> Student Table </h1>  
    <table style="width:50%">  
        <tr>  
            <th>Firstname</th>  
            <th>Lastname</th>  
            <th>Age</th>  
        </tr>  
        <tr>  
            <td>Priya</td>  
            <td>Sharma</td>  
            <td>24</td>  
        </tr>  
        <tr>  
            <td>Arun</td>  
            <td>Singh</td>  
            <td>32</td>  
        </tr>  
        <tr>  
            <td>Sam</td>
```

```
<td>Watson</td>
<td>41</td>
</tr>
</table>
</body>
</html>
```

Student Table

Firstname	Lastname	Age
Priya	Sharma	24
Arun	Singh	32
Sam	Watson	41

Adding Collapsed Borders in an HTML Table

For borders to collapse into one border, add the CSS border-collapse property.

Syntax

```
table, th, td {
    border: 1px solid black;
    border-collapse: collapse;
}
```

```
<!-- index.html -->
<!DOCTYPE html>
<html>

<head>
  <style>
    table,
    th,
    td {
      border: 1px solid black;
      border-collapse: collapse;
    }
  </style>
</head>

<body>
```

```
<h1> Student Table </h1>
<table style="width:50%">
  <tr>
    <th>Firstname</th>
    <th>Lastname</th>
    <th>Age</th>
  </tr>
  <tr>
    <td>Priya</td>
    <td>Sharma</td>
    <td>24</td>
  </tr>
  <tr>
    <td>Arun</td>
    <td>Singh</td>
    <td>32</td>
  </tr>
  <tr>
    <td>Sam</td>
    <td>Watson</td>
    <td>41</td>
  </tr>
</table>
</body>

</html>
```

Student Table

Firstname	Lastname	Age
Priya	Sharma	24
Arun	Singh	32
Sam	Watson	41

Adding Cell Padding in an HTML Table

Cell padding specifies the space between the cell content and its borders. If we do not specify a padding, the table cells will be displayed without padding.

Syntax

```
th, td {  
    padding: 20px;  
}
```

```
<!-- index.html -->  
<!DOCTYPE html>  
<html>  
  
<head>  
    <style>  
        table,  
        th,  
        td {  
            border: 1px solid black;  
            border-collapse: collapse;  
            padding: 20px;  
        }  
    </style>  
</head>  
  
<body>  
    <h1> Student Table </h1>  
    <table style="width:50%">  
        <tr>  
            <th>Firstname</th>  
            <th>Lastname</th>  
            <th>Age</th>  
        </tr>  
        <tr>  
            <td>Priya</td>  
            <td>Sharma</td>  
            <td>24</td>  
        </tr>  
        <tr>  
            <td>Arun</td>  
            <td>Singh</td>  
            <td>32</td>  
        </tr>
```

```
<tr>
  <td>Sam</td>
  <td>Watson</td>
  <td>41</td>
</tr>
</table>
</body>
</html>
```

Student Table

Firstname	Lastname	Age
Priya	Sharma	24
Arun	Singh	32
Sam	Watson	41

Adding Left Align Headings in an HTML Table

By default, the table headings are bold and centered. To left-align the table headings, we must use the CSS text-align property.

Syntax

```
th {
  text-align: left;
}
```

Adding Cells that Span Many Columns in HTML Tables

To make a cell span more than one column, we must use the colspan attribute.

Example: Use of [colspan attribute](#) in HTML.

HTML

```
<!DOCTYPE html>
<html>

<head>
  <style>
    table,
    th,
    td {
      border: 1px solid black;
      border-collapse: collapse;
      padding: 20px;
    }
  </style>
</head>

<body>
  <h2>Cell that spans two columns:</h2>
  <table style="width:50%">
    <tr>
      <th>Name</th>
      <th colspan="2">Telephone</th>
    </tr>
    <tr>
      <td>Vikas Rawat</td>
      <td>9125577854</td>
      <td>8565557785</td>
    </tr>
  </table>
</body>

</html>
```


Cell that spans two columns:

Name	Telephone	
Vikas Rawat	9125577854	8565557785

Adding Cells that span many rows in HTML Tables

To make a cell span more than one row, we must use the rowspan attribute.

Example: Use of the [rowspan attribute](#) in HTML.

```
<!-- index.html -->
<!DOCTYPE html>
<html>

<head>
  <style>
    table,
    th,
    td {
      border: 1px solid black;
      border-collapse: collapse;
      padding: 20px;
    }
  </style>
</head>

<body>
  <h2>Cell that spans two rows:</h2>
  <table style="width:50%">
    <tr>
      <th>Name:</th>
      <td>Vikas Rawat</td>
    </tr>
    <tr>
      <th rowspan="2">Telephone:</th>
      <td>9125577854</td>
    </tr>
    <tr>
      <td>8565557785</td>
    </tr>
```

```
</table>
</body>

</html>
```

Cell that spans two rows:

Name:	Vikas Rawat
Telephone:	9125577854
	8565557785

Adding a *Caption* in an HTML Table

To add a caption to a table, we must use the “caption” tag.

Syntax

```
<table style="width:100%">
<caption>DETAILS</caption>
```

```
<!-- index.html -->
<!DOCTYPE html>
<html>

<head>
  <style>
    table,
    th,
    td {
      border: 1px solid black;
      border-collapse: collapse;
      padding: 20px;
    }
  </style>
</head>

<body>
  <h1 > Student Table </h1>
```

```
<table style="width:50%">
  <caption> Student details</caption>
  <tr>
    <th>Firstname</th>
    <th>Lastname</th>
    <th>Age</th>
  </tr>
  <tr>
    <td>Priya</td>
    <td>Sharma</td>
    <td>24</td>
  </tr>
  <tr>
    <td>Arun</td>
    <td>Singh</td>
    <td>32</td>
  </tr>
  <tr>
    <td>Sam</td>
    <td>Watson</td>
    <td>41</td>
  </tr>
</table>
</body>

</html>
```

Student Table

Student details		
Firstname	Lastname	Age
Priya	Sharma	24
Arun	Singh	32
Sam	Watson	41

Adding a Background Colour to the Table

A color can be added as a background in an HTML table using the “*background-color*” option.

Syntax

```
table#t01 {  
    width: 100%;  
    background-color: #f2f2d1;  
}
```

```
<!-- index.html -->  
<!DOCTYPE html>  
<html>  
  
<head>  
    <style>  
        table,  
        th,  
        td {  
            border: 1px solid black;  
            border-collapse: collapse;  
            padding: 20px;  
        }  
        table#t01 {  
            width: 50%;  
            background-color: red;  
        }  
    </style>  
</head>  
  
<body>  
    <h1> Student Table </h1>  
    <table style="width:50%">  
        <caption> Student details</caption>  
        <table id="t01">  
            <tr>  
                <th>Firstname</th>  
                <th>Lastname</th>  
                <th>Age</th>  
            </tr>  
            <tr>  
                <td>Priya</td>  
                <td>Sharma</td>  
                <td>24</td>
```

```
</tr>
<tr>
  <td>Arun</td>
  <td>Singh</td>
  <td>32</td>
</tr>
<tr>
  <td>Sam</td>
  <td>Watson</td>
  <td>41</td>
</tr>
</table>
</body>
</html>
```

Student Table

Student details

Firstname	Lastname	Age
Priya	Sharma	24
Arun	Singh	32
Sam	Watson	41

HTML Form: An HTML form is a section of a document which contains controls such as text fields, password fields, checkboxes, radio buttons, submit button, menus etc.

An HTML form facilitates the user to enter data that is to be sent to the server for processing such as name, email address, password, phone number, etc.

Why use HTML Form HTML forms are required?

- If you want to collect some data from of the site visitor.

HTML Form Syntax

1. `<form action="server url" method="get|post">`
2. `//input controls e.g. textfield, textarea, radiobutton, button`
3. `</form>`

HTML `<input>` element

The HTML `<input>` element is fundamental form element. It is used to create form fields, to take input from user. We can apply different input field to gather different information from user. Following is the example to show the simple text input.

Example:

1. `<body>`
2. `<form>`
3. Enter your name `<br>`
4. `<input type="text" name="username">`
5. `</form>`
6. `</body>`

Output:

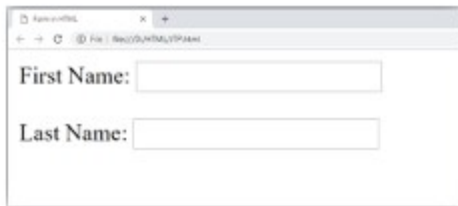
Enter your name

HTML TextField Control

The type="text" attribute of input tag creates textfield control also known as single line textfield control. The name attribute is optional, but it is required for the server side component such as JSP, ASP, PHP etc.

1. `<form>`
2. First Name: `<input type="text" name="firstname"/> <br/>`
3. Last Name: `<input type="text" name="lastname"/> <br/>`
4. `</form>`

Output:



First Name:

Last Name:

Label Tag in Form

It is considered better to have label in form. As it makes the code parser/browser/user friendly.

If you click on the label tag, it will focus on the text control. To do so, you need to have for attribute in label tag that must be same as id attribute of input tag.

1. `<form>`
2. `<label for="firstname">First Name: </label> <br/>`
3. `<input type="text" id="firstname" name="firstname"/> <br/>`
4. `<label for="lastname">Last Name: </label>`
5. `<input type="text" id="lastname" name="lastname"/> <br/>`
6. `</form>`

Output:



HTML Password Field Control

The password is not visible to the user in password field control.

1. `<form>`
2. `<label for="password">Password: </label>`
3. `<input type="password" id="password" name="password"/> <br/>`
4. `</form>`

Output:



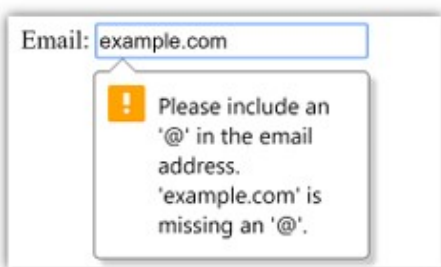
HTML 5 Email Field Control

The email field is new in HTML 5. It validates the text for correct email address. You must use @ and . in this field.

1. `<form>`
2. `<label for="email">Email: </label>`
3. `<input type="email" id="email" name="email"/> <br/>`
4. `</form>`

It will display in browser like below:

Email:



Radio Button Control

The radio button is used to select one option from multiple options. It is used for selection of gender, quiz questions etc.

If you use one name for all the radio buttons, only one radio button can be selected at a time.

Using radio buttons for multiple options, you can only choose a single option at a time.

1. `<form>`
2. `<label for="gender">Gender: </label>`
3. `<input type="radio" id="gender" name="gender" value="male"/>Male`
4. `<input type="radio" id="gender" name="gender" value="female"/>Female <br/>`
5. `</form>`

Gender: ☐ Male ☒ Female

Checkbox Control

The checkbox control is used to check multiple options from given checkboxes.

1. `<form>`
2. Hobby: `<br>`
3. `<input type="checkbox" id="cricket" name="cricket" value="cricket"/>`
4. `<label for="cricket">Cricket</label> <br>`
5. `<input type="checkbox" id="football" name="football" value="football"/>`
6. `<label for="football">Football</label> <br>`
7. `<input type="checkbox" id="hockey" name="hockey" value="hockey"/>`
8. `<label for="hockey">Hockey</label>`
9. `</form>`

Output:

Hobby:
☒ Cricket
☒ Football
☐ Hockey

Submit button control

HTML `<input type="submit">` are used to add a submit button on web page. When user clicks on submit button, then form get submit to the server.

Syntax:

1. `<input type="submit" value="submit">`

The type = submit , specifying that it is a submit button

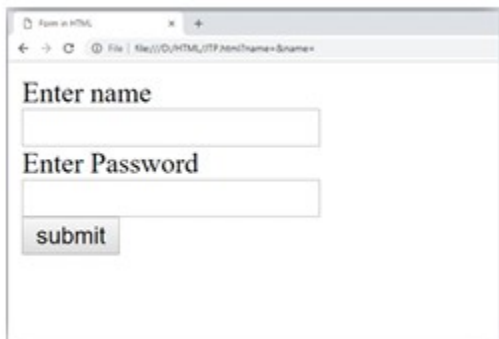
The value attribute can be anything which we write on button on web page.

The name attribute can be omit here.

Example:

1. `<form>`
2. `<label for="name">Enter name</label><br>`
3. `<input type="text" id="name" name="name"><br>`
4. `<label for="pass">Enter Password</label><br>`
5. `<input type="Password" id="pass" name="pass"><br>`
6. `<input type="submit" value="submit">`
7. `</form>`

Output:



HTML <fieldset> element:

The <fieldset> element in HTML is used to group the related information of a form. This element is used with <legend> element which provide caption for the grouped elements.

Example:

1. <form>
2. <fieldset>
3. <legend>User Information:</legend>
4. <label for="name">Enter name</label>

5. <input type="text" id="name" name="name">

6. <label for="pass">Enter Password</label>

7. <input type="Password" id="pass" name="pass">

8. <input type="submit" value="submit">
9. </fieldset>
10. It;/form>

Output:



Example of HTML Form:

```
<!DOCTYPE html>
<html>
<head>
  <title>Simple HTML Form</title>
</head>
<body>
  <h2>Student Registration Form</h2>

  <form action="#" method="post">
    <!-- Label and Text Input -->
    <label for="name">Name:</label>
    <input type="text" id="name" name="name"><br><br>

    <!-- Label and Email Field -->
    <label for="email">Email:</label>
    <input type="email" id="email" name="email"><br><br>

    <!-- Label and Password Field -->
    <label for="password">Password:</label>
    <input type="password" id="password" name="password"><br><br>

    <!-- Radio Buttons -->
    <p>Gender:</p>
    <input type="radio" id="male" name="gender" value="male">
    <label for="male">Male</label><br>
    <input type="radio" id="female" name="gender" value="female">
    <label for="female">Female</label><br><br>

    <!-- Checkboxes -->
    <p>Select your hobbies:</p>
    <input type="checkbox" id="reading" name="hobby1" value="Reading">
    <label for="reading">Reading</label><br>
    <input type="checkbox" id="sports" name="hobby2" value="Sports">
    <label for="sports">Sports</label><br><br>

    <!-- Textarea -->
    <label for="about">About Yourself:</label><br>
    <textarea id="about" name="about" rows="4" cols="40" placeholder="Write something here..."></textarea><br><br>

    <!-- Submit Button -->
```

```
<input type="submit" value="Submit">
</form>
</body>
</html>
```

Student Registration Form

Name:

Email:

Password:

Gender:

- ☐ Male
☐ Female

Select your hobbies:

- ☐ Reading
☐ Sports

About Yourself:

Write something here...

*****END OF UNIT*****

UNIT RELATED QUESTION

Q.1 What do you mean by HTML? Write its features.

Q.2 What is the difference between an HTML element and an HTML attribute & HTML Tags?

Q.3 Explain the purpose of the <head> and <body> & <title> sections in an HTML document.

Q.4 Explain the basic structure of HTML with purpose of each section.

Q.5 Define following HTML formatting tags. For each, write the **tag name**, its **purpose (definition)**, **syntax**, and **one example**: , , <i>, , <mark>, <u>, <strike>, <sup>, , <ins>, <big>, <small>

Q.6 Define the following HTML tags with their syntax and one example for each: <h1> to <h6>, <p>, <a>, , <div>, , , , , <table>, <tr>, <td>, <form>, and <input>,
.

Q.7 Define following HTML List tags. For each, write the **tag name**, its **purpose (definition)**, **syntax**, and **one example**: i) ol, ii) ul, iii) dl, iv) dd, v) dt, vi) li.

Q.8 What is the difference between and tags with example.

Q.9 Write short note on i) Radio Button Control, ii) Password field Control, iii) Check-box Control, iv) Submit Button, v) Textarea, vi) Input element, vii) Label Element.

Q.10 Create an HTML document that displays an ordered list with a minimum of three list items.

Q.11 Create an HTML document that displays an un-ordered list with a minimum of three list items.

Q.12 Define following HTML Table styling property. For each, write, the **purpose (definition)**, **syntax**, and **one example**: i) Adding a border to HTML Table, ii) Adding a collapsed border to HTML Table, iii) Adding a cell padding to HTML Table, iv) Adding a cells that span many rows to HTML Table, v) Adding a cells that span many columns to HTML Table, vi) Adding a caption to HTML Table.

Q.13 Write an HTML program that demonstrates the use of at least five different text formatting tags.

Q.14 Write HTML code to generate a table with 3 rows and 3 columns, including custom data in each cell.

Q.15 Write an HTML code to create a registration form with fields for name, mobile number, password, email ID, gender (radio buttons), hobbies (checkboxes), a 50-word textarea, and a submit button.

UNIT-3

Cascading Style Sheets

- Concept of CSS, Creating CSS,
- CSS properties,
- CSS styling (Background, Text Format, Controlling Fonts),
- Working with block elements ,
- Working with lists and tables,
- CSS id and class,
- Box Model (Introduction, Border properties, Padding Properties, Margin properties).
- CSS Color,
- Page Layout and Site Designs

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CSS Introduction

CSS (Cascading Style Sheets) is a simply designed language intended to simplify the process of making web pages presentable. CSS allows you to apply styles to HTML documents. It describes how a webpage should look. It prescribes colors, fonts, spacing, etc. In short, you can make your website look however you want. CSS lets developers and designers define how it behaves, including how elements are positioned in the browser. HTML uses tags and CSS uses rule sets. CSS styles are applied to the HTML element using selectors. CSS is easy to learn and understand, but it provides powerful control over the presentation of an HTML document

Why CSS?

- **CSS saves time:** You can write CSS once and reuse the same sheet in multiple HTML pages.
- **Easy Maintenance:** To make a global change simply change the style, and all elements in all the WebPages will be updated automatically.
- **Search Engines:** CSS is considered a clean coding technique, which means search engines won't have to struggle to "read" its content.
- **Superior styles to HTML:** CSS has a much wider array of attributes than HTML, so you can give a far better look to your HTML page in comparison to HTML attributes.
- **Offline Browsing:** CSS can store web applications locally with the help of an offline cache. Using this we can view offline websites.

CSS Syntax

CSS (Cascading Style Sheets) is used to style HTML elements.

The basic syntax is:

```
// CSS Style
h1 { color: blue; font-size: 12px; }

Where -
Selector - h1
Declaration - { color: blue; font-size: 12px; }
```

- The selector points to the HTML element that you want to style.
- The declaration block contains one or more declarations separated by semicolons.
- Each declaration includes a CSS property name and a value, separated by a colon.

There are 3 ways to apply CSS to your HTML:

Method	Description	Where to Write
Inline CSS	Directly inside HTML tags	Inside the style attribute
Internal CSS	Within a <style> tag inside the <head>	Inside the HTML file
External CSS	In a separate .css file	Linked to HTML

➤ **Inline CSS:**

```
html
<h1 style="color: red;">Hello World</h1>
```

➤ **Internal CSS:**

```
html
<head>
  <style>
    h1 {
      color: green;
    }
  </style>
</head>
```

➤ **External CSS:**

- In your HTML file:

```
html
<link rel="stylesheet" href="styles.css">
```

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- In styles.css:

```
css
h1 {
  color: purple;
}
```

CSS (Cascading Style Sheets) properties

CSS (Cascading Style Sheets) properties are used to style HTML elements on web pages. They allow you to control the appearance, layout, and behavior of elements.

Property	Description	Example
color	Sets the color of the text	color: red;
background-color	Sets the background color	background-color: lightblue;
font-family	Specifies the font style for the text	font-family: Arial, sans-serif;
font-size	Sets the size of the text	font-size: 16px;
font-weight	Controls the boldness/thickness of text	font-weight: bold;
text-decoration	Adds text styles like underline, overline, etc.	text-decoration: underline;
margin	Sets the space outside the element	margin: 10px; padding: 5px;
padding	Sets the space inside the	border: 1px solid black;
border	Adds a border around	width: 200px;
width	Sets the height of an element	height: 100px;
display	Specifies how the element	display: block;

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CSS Background

The CSS background properties are used to define the background effects for elements like color, image, position, size, and more.

They help you customize how an element looks behind its content.

CSS Background – Syntax

```
selector {
  background-property: value;
}
```

CSS Background Properties

Property	Definition	Syntax / Concept	Example
background-color	Sets the background color of an element.	background-color: color;	body { background-color: green; }
background-image	Sets an image as the background of an element.	background-image: url('image.jpg');	body { background-image: url('bg.jpg'); }
background-repeat	Controls if/how background image repeats.	background-repeat: repeat / no-repeat;	body { background-repeat: no-repeat; }
background-attachment	Fixes background image while scrolling.	background-attachment: fixed;	body { background-attachment: fixed; }
background-position	Sets position of the background image.	background-position: top left / center;	body { background-position: center; }

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CSS Text Formatting Properties

Property	Definition	Example
color	Sets text color using name, hex, or RGB.	color: red; or color: #ff0000;
text-align	Aligns text (left, right, center, justify).	text-align: center;
text-align-last	Aligns only the last line of text.	text-align-last: right;
text-decoration	Adds decoration (underline, overline, line-through).	text-decoration: underline;
text-decoration-color	Sets color of the text decoration.	text-decoration-color: blue;
text-decoration-line	Type of line decoration.	text-decoration-line: overline;
text-decoration-style	Style of the decoration line (solid, dotted, etc.).	text-decoration-style: wavy;
text-indent	Indents the first line of a paragraph.	text-indent: 30px;
text-justify	Sets justification method for aligned text.	text-justify: inter-word;
text-overflow	Controls overflowed text (ellipsis, clip).	text-overflow: ellipsis;
text-transform	Changes letter case (uppercase, lowercase).	text-transform: uppercase;
text-shadow	Adds shadow effect to text.	text-shadow: 2px 2px 5px gray;
letter-spacing	Adjusts space between characters.	letter-spacing: 2px;
line-height	Sets space between lines.	line-height: 1.5;
direction	Sets text direction (ltr or rtl).	direction: rtl;
word-spacing	Sets space between words.	word-spacing: 10px;

CSS Fonts Properties

Property	Description	Syntax	Example
font-family	Specifies the font type of an element.	font-family: "font family name";	font-family: "Arial", sans-serif;
font-style	Specifies the font style (normal, italic, oblique).	font-style: style name;	font-style: italic;
font-weight	Sets the weight/thickness of the font.	font-weight: font weight value;	font-weight: bold;
font-variant	Creates a small-caps effect.	font-variant: font variant value;	font-variant: small-caps;
font-size	Sets the font size using px, %, em, etc.	font-size: font size value;	font-size: 16px;
font-stretch	Sets the text wider or narrower.	font-stretch: value;	font-stretch: expanded;
font-kerning	Controls the usage of kerning info in the font.	font-kerning: normal none auto;	font-kerning: normal;

Example: HTML Code applying all CSS Background & CSS Properties.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Font Styling Example</title>
  <style>
    h1 {
      font-size: 36px; /* Set the font size to 36 pixels */
      font-variant: normal; /* Normal variant (no small-caps) */
      font-weight: bold; /* Bold text */
      font-style: italic; /* Italic text */
      font-family: 'Courier New', monospace; /* Monospace font family */
    }
  </style>
</head>
<body>
  <h1>This is a Styled Heading</h1>
```

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```
<p>This is a single example of CSS fonts.</p>
</body>
</html>
```

This text has a background color.

This text has a background image that does not scroll.

This text is green.

This text is center aligned.

This text is justified, but the last line is right aligned.

This text has dotted red underline decoration.

This paragraph has an indented first line.

This paragraph is justified using "inter-word".

This is a long line that ...

THIS TEXT IS TRANSFORMED TO UPPERCASE.

This text has a shadow.

Spaced letters using letter-spacing.

This text has double line height.

Each line has more space in between.

Block Elements

The elements that begin on a new line are known as block elements. A block element acquires up the full width available for that content. Unlike inline, there exists a top and bottom margin for these elements. Block-level elements may only appear inside the body tag. Block-level elements create a larger structure than inline elements.

Example of block elements:

<div>, <p>, , <main>, <nav>, , <form>, <video>, <table>, <aside>, <article> are some of the inline tags.

```
<!DOCTYPE html>
<html>
<head>
```

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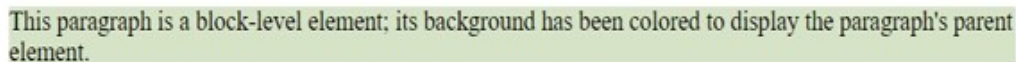
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```
<title>Block Element Example</title>
<style>
  p {
    background-color: #88aaff; /* Valid HEX color */
    padding: 10px;
  }
</style>
</head>
<body>

  <p>This paragraph is a block-level element. It starts on a new line and spans the
  full width.</p>

</body>
</html>
```

Output:



This paragraph is a block-level element; its background has been colored to display the paragraph's parent element.

Inline Block Elements

The display value of inline-block works similarly to inline with one exception: the height and width of that element become modifiable.

```
<!DOCTYPE html>
<html>
<head>
  <title>Inline-Block Element Example</title>
  <style>
    span.b {
      display: inline-block; /* Makes the span behave like a block but inline */
      width: 100px;         /* Set the width of the element */
      height: 100px;        /* Set the height of the element */
      padding: 5px;         /* Add padding inside the span */
      color: white;          /* Text color inside the span */
      border: 1px solid blue; /* Border around the span */
    }
  </style>
</head>
<body>
  <span class="b">This is an inline-block element.</span>
</body>
</html>
```


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```
background-color: blue; /* Background color of the span */
text-align: center; /* Centers the text horizontally */
line-height: 90px; /* Vertically centers the text within the span */
}
</style>
</head>
<body>

<div>This is an example of an inline-block element with a span colored as blue.
  <span class="b">Inline Block</span> is an inline-block element.
</div>

</body>
</html>
```

Output

This is an example of an inline-block element with a span colored as blue. **Inline Block** is an inline-bl

Example: A complete code including all these displays property.

```
<!DOCTYPE html>
<html>
<head>
  <title>Display Property Example</title>
  <style>
    /* Inline display property */
    span.a {
      display: inline; /* The default for span */
      width: 100px; /* width and height don't apply to inline elements */
      height: 100px;
      padding: 5px;
      border: 1px solid white;
      background-color: blue;
    }

    /* Inline-block display property */
```

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```
span.b {
    display: inline-block; /* The element behaves like a block element */
    width: 100px;
    height: 100px;
    padding: 5px;
    border: 1px solid white;
    background-color: green;
}

/* Block display property */
span.c {
    display: block; /* Makes the span behave like a block element */
    width: 70px;
    height: 50px;
    padding: 5px;
    border: 1px solid blue;
    background-color: pink;
}
</style>
</head>
<body>

<h1>The Display Property</h1>

<h2>display: inline</h2>
<div>This is a section with an inline span made blue:
    <span class="a">inline</span>
    <span class="a">element</span>
</div>

<h2>display: block</h2>
<div>This is a section with a block span made pink:
    <span class="c">block</span>
    <span class="c">element</span>.
</div>

<h2>display: inline-block</h2>
<div>This is a section with an inline-block span made green:
    <span class="b">inline</span>
    <span class="b">block</span>
</div>

</body>
</html>
```

Output:

The Display Property

display: inline

This is a section with an inline span made blue: inline element

display: block

This is a section with a block span made pink:

block

element

display: inline-block

This is a section with an inline-block span made green: inline block

Difference between Inline, Block and Inline -Block

Inline Example	Block Example	Inline-Block Example
It does not start on a new line; flows within the same line	It starts on a new line; takes full width	It does not start on a new line; behaves like inline but accepts width and height
It cannot set width and height	It can set width and	It can set width and height

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Inline Example	Block Example	Inline-Block Example
	height	
Common HTML Tags are span, a, strong, em	Common HTML Tags are div, p, h1–h6, ul, li	Common HTML Tags are button, input, custom divs styled with display: inline-block
CSS Display Declaration is display: inline;	CSS Display Declaration is display: block;	CSS Display Declaration is display: inline-block;

CSS Lists

The **List** in CSS specifies the listing of the contents or items in a particular manner i.e., it can either be organized orderly or unordered way, which helps to make a clean webpage. It can be used to arrange the huge with a variety of content as they are flexible and easy to manage. The default style for the list is borderless.

1. Item 1
2. Item 2
3. Item 3

Currently Active Property:

list-style-type: decimal;

The list can be categorized into 2 types:

- **Unordered List:** In unordered lists, the list items are marked with bullets i.e. small black circles by default.
- **Ordered List:** In ordered lists, the list items are marked with numbers and an alphabet.

We have the following CSS lists properties, which can be used to control the CSS lists:

- **list-style-type:** This property is used to specify the appearance (such as disc, character, or custom counter style) of the list item marker.
- **list-style-image:** This property is used to set the images that will be used as the list item marker.
- **list-style-position:** It specifies the position of the marker box with respect to the principal block box.
- **list-style:** This property is used to set the list style.

Now, we will learn more about these properties with examples.

List Item Marker

This property specifies the type of item marker i.e. unordered list or ordered. The **list-style-type** property specifies the appearance of the list item marker (such as a disc, character, or custom counter style) of a list item element. Its default value is a disc.

Syntax:

list-style-type: value;

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CSS list-style-type Values

CSS List Style Type	Description
none	No marker or bullet is displayed.
circle	Default marker for unordered lists, displays a hollow circle (○).
square	Displays filled square markers (■) for unordered lists.
decimal	Default marker for ordered lists: 1, 2, 3, ...
decimal-leading-zero	Displays decimal numbers with leading zeroes: 01, 02, 03, ...
lower-roman	Displays lowercase Roman numerals: i, ii, iii, ...
upper-roman	Displays uppercase Roman numerals: I, II, III, ...
lower-alpha	Displays lowercase letters: a, b, c, ...
upper-alpha	Displays uppercase letters: A, B, C, ...

Example: HTML CSS Code for CSS List Style

```
<!DOCTYPE html>
<html>

<head>
  <title>CSS List Styling Example</title>
  <style>
    ul.a {
      list-style: square;
      background: pink;
      padding: 20px;
    }

    ul.b {
      list-style-type: circle;
    }

    ol.c {
      list-style-type: lower-alpha;
    }

    ol.d {
      list-style-type: upper-roman;
    }
  </style>
</head>
```

```
<body>
  <h2>GeeksforGeeks</h2>

  <p>Unordered Lists</p>
  <ul class="a">
    <li>one</li>
    <li>two</li>
    <li>three</li>
  </ul>

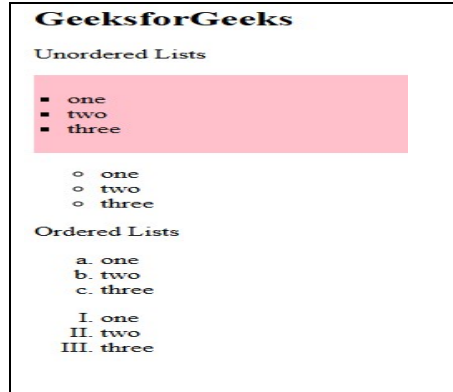
  <ul class="b">
    <li>one</li>
    <li>two</li>
    <li>three</li>
  </ul>

  <p>Ordered Lists</p>
  <ol class="c">
    <li>one</li>
    <li>two</li>
    <li>three</li>
  </ol>

  <ol class="d">
    <li>one</li>
    <li>two</li>
    <li>three</li>
  </ol>
</body>

</html>
```

Output:



CSS id and class

In CSS (Cascading Style Sheets), both IDs and classes are used to apply styles to HTML elements, but they have different purposes and specificity.

1. **ID Selector:** An ID selector is denoted by the # symbol followed by an identifier. IDs are unique within a webpage, meaning an ID should only be assigned to one element on a page. They are often used to style a specific element that will not be repeated elsewhere on the page.

Example:

```
#uniqueElement { color: blue; }
```

In HTML:

```
<div id="uniqueElement">This element has a unique style.</div>
```

2. **Class Selector:** A class selector is denoted by a period (.) followed by an identifier. Classes are not unique and can be applied to multiple elements on a page. They are commonly used to style groups of elements that share similar characteristics.

Example:

```
.button { background-color: #007bff; color: white; padding: 10px 20px; border: none; border-radius: 5px; }
```

In HTML:

```
<button class="button">Click me</button> <button class="button">Submit</button>
```


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Box Model (Introduction, Border properties, Padding Properties, Margin properties)-

The CSS box model is a fundamental concept that describes how elements on a web page are structured and how they interact with each other. It consists of four main components: content, padding, border, and margin.

1. **Content:** The actual content of the HTML element, such as text, images, or other media.
2. **Padding:** The space between the content and the element's border. Padding helps create space around the content, separating it from the border.
3. **Border:** The border that surrounds the padding and content of the element. It can be styled using various properties such as color, width, and style.
4. **Margin:** The space outside the border of the element. Margins are used to create space between elements, controlling their layout and spacing on the page.

Let's explore some CSS properties related to each component:

Border Properties:

- **border-width:** Specifies the width of the border.
- **border-style:** Specifies the style of the border (e.g., solid, dashed, dotted).
- **border-color:** Specifies the color of the border.
- **border:** Shorthand property for setting the width, style, and color of the border in one declaration.

CSS Box Model: Padding and Margin

Every HTML element is wrapped in a “box” whose inner and outer spacing you can control.

Padding

Padding is the space between an element's content and its border. You can set each side individually or all at once:

- **padding-top, padding-right, padding-bottom, padding-left:** Specify the padding on each side in units such as pixels (px), ems (em), or percentages.
- **padding:** Shorthand that sets the top, right, bottom, and left padding in one declaration (e.g. padding: 10px 20px 10px 20px;).

Margin

Margin is the space outside an element's border, separating it from neighboring elements. Similarly, you can set margins per side or altogether:

- **margin-top, margin-right, margin-bottom, margin-left:** Control the external spacing on each side individually.
- **margin:** Shorthand to define all four margins at once (e.g. margin: 5px 10px; which applies 5px top/bottom and 10px left/right).

CSS Color Specification

CSS lets you choose colors in five main ways:

1. Predefined Color Names

There are 140 built-in names (such as red, blue, green, teal, goldenrod). Using a name is as simple as: color: navy; background-color: lightgray;

2. Hexadecimal Codes

A hex code begins with # followed by six hexadecimal digits. Each pair represents the red, green, and blue channels: color: #FF0000; /* pure red */ background-color: #00FF00; /* pure green */

3. RGB Function

The rgb() function takes three numbers (0–255) for red, green, and blue intensities: color: rgb(255, 0, 0); /* red */

4. HSL Function

The hsl() function describes color by hue (0–360°), saturation (0–100%), and lightness (0–100%): color: hsl(120, 100%, 50%); /* bright green */

5. RGBA Function

Like rgb(), but adds an alpha channel (0–1) for opacity: background-color: rgba(0, 0, 255, 0.5); /* semitransparent blue */

Page Layout & Site Design (HTML & CSS)

1. HTML Structural Elements

- **header:** Top of page—logo, site title, main navigation.
- **nav:** Collection of links for site navigation.
- **main:** Core content of the page.
- **section / article:** Thematic grouping of related content.
- **aside:** Sidebar—related info, ads, secondary links.
- **footer:** Bottom of page—contact info, copyright.

2. CSS Box Model

Each element is a rectangular box composed of:

- Content (text, images)
- Padding (inside spacing)
- Border (edge)
- Margin (outside spacing)

Use padding and margin (individual sides or shorthand) to control whitespace.

3. Layout Techniques

- **Flexbox (display: flex;):** One-dimensional layout (row or column).
- **Grid (display: grid;):** Two-dimensional layout (rows × columns).
- **Float:** Older method to wrap text—largely superseded by flex and grid.
- **Positioning:**
 - static (default)
 - relative (offset from normal)
 - absolute (placed relative to nearest positioned ancestor)
 - fixed (viewport-fixed)
 - sticky (toggles between relative/fixed).

4. Responsive Design

- **Viewport meta:** It gives the browser instruction on how to control the page's dimensions & scaling.

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

- **Media queries adapt layouts at breakpoints:**

```
@media (max-width: 600px) {  
  /* mobile styles */  
}
```

5. Visual Design Principles

- **Consistent spacing:** Uniform margins/padding.
- **Limited color palette:** Choose 2–3 harmonious colors.
- **Readable typography:** Clear font-families, comfortable line-height.
- **Visual hierarchy:** Distinguish headings, subheadings, body text.
- **Whitespace:** Avoid clutter—let elements breathe.

Example: Page Layout of HTML & CSS

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8">  
  <meta name="viewport" content="width=device-width, initial-scale=1.0">  
  <title>Layout & Design Demo</title>  
  
  <style>  
    /* —— Global & Typography —— */  
    /* Define CSS custom properties (variables) for colors & spacing */  
    :root {  
      --primary: #2a9d8f; /* main accent color */  
      --accent: #e9c46a; /* secondary accent color */  
      --light: #f4f4f9; /* page background */  
      --dark: #264653; /* primary text color */  
      --gap: 16px; /* consistent spacing unit */  
    }  
  
    /* Apply border-box model universally */  
    * {  
      box-sizing: border-box; /* include padding/border in element's width/height */  
      margin: 0;  
      padding: 0;  
    }  
  
    body {  
      font-family: 'Segoe UI', Tahoma, sans-serif; /* readable font stack */  
      line-height: 1.6; /* comfortable line spacing */  
      color: var(--dark); /* text color from variable */
```

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```
background: var(--light);          /* page background */
padding: var(--gap);               /* outer page padding */
}

h1, h2 {
  margin-bottom: var(--gap); /* space below headings */
}
p {
  margin-bottom: var(--gap); /* space below paragraphs */
}

/* _____ Page Structure _____ */
/* Style semantic sections uniformly */
header, nav, main, aside, footer {
  padding: var(--gap);
  background: white;
  border: 1px solid #ccc;
  margin-bottom: var(--gap);
}

/* Header specific styling */
header {
  background: var(--primary);
  color: white;
}

/* Navigation menu */
nav ul {
  list-style: none;    /* remove default bullets */
  display: flex;       /* horizontal menu */
  gap: var(--gap);     /* space between links */
}
nav a {
  text-decoration: none; /* remove underline */
  color: white;          /* link color */
}

/* _____ Flex Layout _____ */
.flex-container {
  display: flex; /* establish flex formatting context */
  gap: var(--gap);
}
.flex-item {
  flex: 1;      /* items share available space equally */
}
```

```
padding: var(--gap);
background: var(--accent);
}

/* —— Grid Layout —— */
.grid-container {
display: grid;
/* create responsive columns: auto-fit as many 150px columns as will fit */
grid-template-columns: repeat(auto-fit, minmax(150px, 1fr));
gap: var(--gap);
}
.grid-item {
padding: var(--gap);
background: var(--primary);
color: white;
}

/* —— Responsive —— */
/* At viewport widths ≤ 600px, stack flex items vertically */
@media (max-width: 600px) {
nav ul {
flex-direction: column;
}
.flex-container {
flex-direction: column;
}
}
</style>
</head>
<body>

<!-- Page header with site title -->
<header>
<h1>My Website</h1>
</header>

<!-- Main navigation bar -->
<nav>
<ul>
<li><a href="#">Home</a></li>
<li><a href="#">About</a></li>
<li><a href="#">Contact</a></li>
</ul>
</nav>
```

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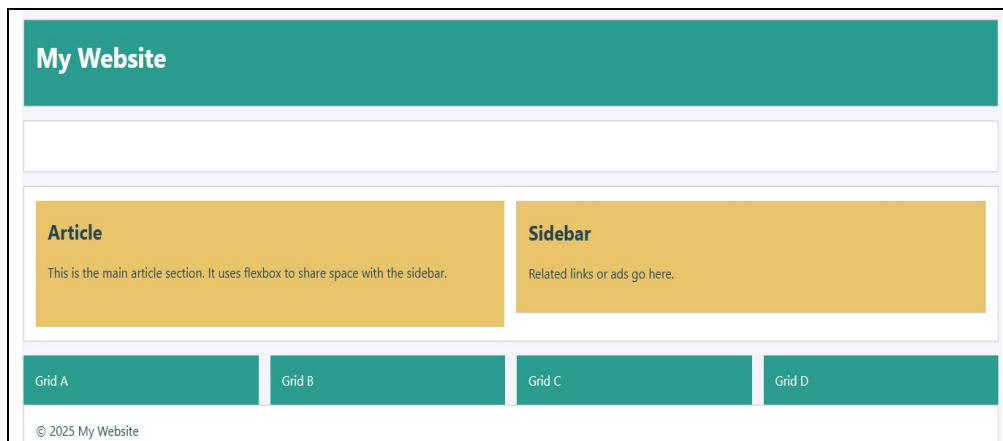
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```
<!-- Flex container: main content and sidebar -->
<main class="flex-container">
  <section class="flex-item">
    <h2>Article</h2>
    <p>This is the main article section. It uses flexbox to share space with the sidebar.</p>
  </section>
  <aside class="flex-item">
    <h2>Sidebar</h2>
    <p>Related links or ads go here.</p>
  </aside>
</main>

<!-- Grid layout: responsive cards -->
<section class="grid-container">
  <div class="grid-item">Grid A</div>
  <div class="grid-item">Grid B</div>
  <div class="grid-item">Grid C</div>
  <div class="grid-item">Grid D</div>
</section>

<!-- Page footer -->
<footer>
  <p>&copy; 2025 My Website</p>
</footer>

</body>
</html>
```



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UNIT RELATED QUESTION

Q.1 What are the three different ways to apply CSS to an HTML document, and how does each method work?

Q.2 What is CSS and why is it used in web development?

Q.3 Write the definition, syntax and example of following properties: background- color, background- image, background-repeat, background – attachment, background- position.

Q.4 Write any four CSS text formatting properties with definition, syntax and example.

Q.5 Write any four CSS font properties with definition, syntax and example.

Q.6 Write an HTML code example that applies any four CSS background and font properties.

Q.7 Write a short note on each i) Block Element, ii) Inline element, and iii) Block – inline element.

Q.8 Write the differences between (i) Block elements, (ii) Inline elements, and (iii) Inline-block elements.

Q.9 What is a CSS list, how many types of lists are there in HTML, and what are the main CSS list properties along with their definitions?

Q.10 Define list item marker with its syntax.

Q.11 Write an HTML document with internal CSS that demonstrates the use of different list-style types for both ordered and unordered lists.

Q.12 What do you mean by Id selector ?

Q.13 What do you mean by class selector?

Q.14 Mention any 3 border properties.

Q.15 What are the three main ways to specify colors in CSS?

Q.16 Write a short note on CSS Box Model(Padding and Margin).

UNIT-4

Java Script

- Introduction to Java script,
- Basics, Variables,
- String manipulation,
- Mathematical Functions,
- Operations,
- Arrays,
- Functions,
- Objects in Java script- regular expressions, Built-in objects,
- Data validation, Messages & Confirmation

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1. JavaScript

JavaScript is a lightweight, cross-platform, single-threaded, and interpreted compiled programming language. It is also known as the scripting language for webpage. It is well-known for the development of web pages, and many non browser environments also use it.

JavaScript is a **weakly typed language (dynamically typed)**. JavaScript can be used for **Client-side developments as well as Server-side developments**. JavaScript is both an **imperative and declarative type of language**. JavaScript contains a standard library of objects, like Array, Date, and Math, and a core set of language elements like operators, control structures, and statements.

Key features of Java Script:

- **Client-side:** It supplies objects to control a browser and it's Document Object Model (DOM). Like if client-side extensions allow an application to place elements on an HTML form and respond to user events such as mouse clicks, form input, and page navigation. Useful libraries for the client side are AngularJS, ReactJS, VueJS, and so many others.
- **Server-side:** It supplies objects relevant to running JavaScript on a server. For if the server-side extensions allow an application to communicate with a database, and provide continuity of information from one invocation to another of the application, or perform file manipulations on a server. The useful framework which is the most famous these days is node.js.
- **Imperative language:** In this type of language we are mostly concerned about how it is to be done. It simply controls the flow of computation. The procedural programming approach, object, oriented approach comes under this as async await we are thinking about what is to be done further after the async call.
- **Declarative programming:** In this type of language we are concerned about how it is to be done, basically here logical computation requires. The main goal is to describe the desired result without direct dictate on on how to get it as the arrow function does.

2. How to Link a JavaScript File in HTML

JavaScript can be added to an HTML file in two main ways:

1. Internal JavaScript: You can write JavaScript code directly within your HTML file using the `<script>` tag. This tag can be placed either in the `<head>` or `<body>` section of the HTML, depending on when you want the script to run.

Syntax:

```
<script>
```


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Welcome to our first program

3. Variables in JavaScript: In JavaScript, **variables** are used to store data values. You can think of a variable as a **container that holds information** which can be used or modified later in the program.

Declaring Variables: Variables in JavaScript can be declared using one of three keywords: **var**, **let**, or **const**.

1. var

- Historically used in JavaScript for declaring variables.
- Has **function scope** or **global scope**, but **not block scope**.
- Can be re-declared and updated.

```
var x = 5;
```

2. let

- Has **block scope**, meaning it's only accessible within the block where it's defined.
- Can be updated but **cannot be re-declared** in the same scope.

```
let y = 10;
```

3. const

- Used to declare **constants** (values that cannot be changed).
- Has **block scope**, cannot be updated or re-declared.

```
const PI = 3.14;
```

Variable Naming Rules

- Variable names **must begin** with a **letter**, **underscore _**, or **dollar sign \$**.
- Following characters can be **letters**, **digits**, **underscores**, or **dollar signs**.
- JavaScript variable names are **case-sensitive** (e.g., myVariable $\neq$ MyVariable).

4. Data Types in JavaScript

JavaScript variables can hold various types of data, such as:

- Strings: "Hello"
- Numbers: 100, 3.14
- Booleans: true, false
- Objects
- Arrays
- Null and Undefined
- Functions

5. Comparison of var, let, and const

Property	var	let	const
Scope	Function-scoped	Block-scoped	Block-scoped
Updating	Mutable	Mutable	Immutable
Redeclare	Allowed	Not allowed	Not allowed
Browser Support	Supported in older browsers	Not supported in older browsers	Not supported in older browsers

6. String Manipulation in JavaScript

String manipulation refers to modifying or transforming string data in various ways. JavaScript provides a wide range of built-in methods to work with strings efficiently. Below are some commonly used string operations:

1. Concatenation: You can combine two or more strings using the + operator or the concat() method.

Examples:

```
let str1 = "Hello, ";
let str2 = "world!";
let result = str1 + str2;    // Using + operator
console.log(result);        // Output: Hello, world!

let combined = str1.concat(str2); // Using concat() method
console.log(combined);        // Output: Hello, world!
```

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2. String Length: Use the .length property to find the number of characters in a string.

```
let message = "JavaScript";  
console.log(message.length); // Output: 10
```

3. Accessing Characters: You can access individual characters in a string using bracket notation or the charAt() method.

```
let str = "Hello";  
console.log(str[0]);    // Output: H  
console.log(str.charAt(1)); // Output: e
```

Note: String indices start at 0.

4. Extracting Substrings: Use substring(), slice(), or substr() to extract parts of a string.

```
let text = "JavaScript Programming";  
console.log(text.substring(0, 10)); // Output: JavaScript  
console.log(text.slice(11));      // Output: Programming
```

5. Changing Case: Convert strings to uppercase or lowercase using toUpperCase() and toLowerCase() methods.

```
let greeting = "Hello World";  
console.log(greeting.toUpperCase()); // Output: HELLO WORLD  
console.log(greeting.toLowerCase()); // Output: hello world
```

6. Finding Substrings: Use indexOf() or lastIndexOf() to find the position of a substring within a string.

```
let sentence = "The quick brown fox jumps over the lazy dog";  
console.log(sentence.indexOf("fox")); // Output: 16  
console.log(sentence.lastIndexOf("the")); // Output: 31
```


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7. Replacing Substrings: Use `replace()` to replace part of a string with another string.

```
let msg = "I love JavaScript";
let newMsg = msg.replace("JavaScript", "coding");
console.log(newMsg); // Output: I love coding
```

7. Mathematical Functions in JavaScript

Function	Description	Example
<code>Math.abs(x)</code>	Returns the absolute value of x	<code>Math.abs(-5);</code> // Returns 5
<code>Math.sqrt(x)</code>	Returns the square root of x	<code>Math.sqrt(16);</code> // Returns 4
<code>Math.pow(x, y)</code>	Returns x raised to the power of y	<code>Math.pow(2, 3);</code> // Returns 8
<code>Math.floor(x)</code>	Returns the largest integer less than or equal to x	<code>Math.floor(4.9);</code> // Returns 4
<code>Math.ceil(x)</code>	Returns the smallest integer greater than or equal to x	<code>Math.ceil(4.1);</code> // Returns 5
<code>Math.round(x)</code>	Rounds x to the nearest integer	<code>Math.round(4.4);</code> // Returns 4
<code>Math.min(...)</code>	Returns the smallest value	<code>Math.min(3, 5, 1);</code> // Returns 1
<code>Math.max(...)</code>	Returns the largest value	<code>Math.max(3, 5, 1);</code> // Returns 5
<code>Math.random()</code>	Returns a random number between 0 and 1	<code>Math.random();</code>
Trigonometric Functions	Compute sine, cosine, tangent of angles (in radians)	<code>Math.sin(Math.PI / 2);</code> // Returns 1

8. Arrays

Arrays in JavaScript are used to store multiple values in a single variable.

- **Creating an Array:**
let fruits = ['apple', 'banana', 'orange'];
- **Accessing Elements:**
console.log(fruits[0]); // Output: 'apple'
- **Modifying Elements:**
fruits[1] = 'grape';
- **Array Methods:**
fruits.push('kiwi'); // Add to end
fruits.pop(); // Remove last
fruits.length; // Array length

9. Functions in JavaScript

Functions in JavaScript are reusable blocks of code designed to perform specific tasks. They allow you to organize, reuse, and modularize code. It can take inputs, perform actions, and return outputs.

```
function sum(x, y) {  
    return x + y;  
}  
console.log(sum(6, 9));
```

Output

15

10. Function Syntax and Working

A function definition is sometimes also termed a function declaration or function statement. Below are the rules for creating a function in JavaScript:

- Begin with the keyword **function** followed by,
- A user-defined function name (In the above example, the name is **sum**)
- A list of parameters enclosed within parentheses and separated by commas (In the above example, parameters are **x** and **y**)
- A list of statements composing the body of the function enclosed within curly braces {} (In the above example, the statement is “return x + y”).

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Return Statement

In some situations, we want to return some values from a function after performing some operations. In such cases, we make use of the return. This is an optional statement. In the above function, “sum()” returns the sum of two as a result.

Function Parameters

Parameters are input passed to a function. In the above example, sum() takes two parameters, x and y.

Calling Functions

After defining a function, the next step is to call them to make use of the function. We can call a function by using the function name separated by the value of parameters enclosed between the parenthesis.

```
// Function Definition
function welcomeMsg(name) {
    return ("Hello " + name + " welcome to Renaissance");
}

let nameVal = "Student";

// calling the function
console.log(welcomeMsg(nameVal));
```

Output

```
Hello Student welcome to Renaissance
```

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11. Objects in JavaScript

In JavaScript, **objects** are collections of key-value pairs used to store and manage data. Almost everything in JavaScript is an object, including arrays, functions, and even strings in many cases.

11.1 Regular Expressions (RegExp Object) in JavaScript

S.No.	Feature	Description	Syntax	Example
1	RegExp Literal	Creates a regular expression using literal syntax	/pattern/modifiers	let regex = /abc/i;
2	RegExp Constructor	Creates a regular expression using constructor function	new RegExp('pattern', 'modifiers')	let regex = new RegExp('abc', 'i');
3	.test()	Tests whether a match is found in a string	regex.test(string)	/hello/.test('hello world'); // true
4	.exec()	Executes a search for a match in a string	regex.exec(string)	/hello/.exec('hello world'); // ['hello']
5	g Modifier	Global search (does not return after first match)	pattern with /g	/abc/g
6	i Modifier	Case-insensitive search	pattern with /i	/abc/i
7	m Modifier	Multi-line search	pattern with /m	/^abc/m

11.2 Built-in Objects in JavaScript

JavaScript provides several **built-in objects** for handling common tasks.

Object	Description	Example
Math	Provides basic mathematical operations.	Math.sqrt(16); // 4
Date	Used to work with dates and times.	new Date();
String	Provides string manipulation methods.	"hello".toUpperCase();
Number	Wrapper object for numeric values.	let num = new Number(10);
Boolean	Wrapper object for boolean values.	new Boolean(false);
Array	Used to store multiple values in a single variable.	let arr = [1, 2, 3];
RegExp	Enables regular expression pattern matching.	/abc/.test("abc"); // true
JSON	Used for parsing and stringifying JSON data.	JSON.parse('{"key":"value"}');
Error	Used for error handling and creating custom errors.	throw new Error("Something went wrong");

12. Data Validation, Messages & Confirmation in JavaScript

1. Data Validation:

Data validation ensures that user inputs are correct and meaningful before they are processed. JavaScript can be used to validate forms in real-time (client-side), improving user experience and reducing server load. Common validation tasks include checking if required fields are filled, ensuring email format is correct, verifying number ranges, etc.

Example:

```
if (document.getElementById("email").value === "") {  
    alert("Email field cannot be empty.");  
}
```

2. Messages (Alerts):

JavaScript provides simple popup messages to notify users. The alert() method displays an information box with an OK button.

Example:

```
alert("Form submitted successfully!");
```

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3. Confirmation Boxes:

The confirm() method displays a message box with OK and Cancel buttons, commonly used to get user confirmation before proceeding.

Example:

```
if (confirm("Are you sure you want to delete this item?")) {  
    // proceed with deletion  
}
```

These features are essential for creating interactive, user-friendly, and secure web applications.

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UNIT RELATED QUESTION

Q.1 Define: i) Java Script, ii) String Concatenation, iii) Imperative Language, iv) Declarative Language, v) Return Statement, vi) Function Parameter, vii) Variable in JavaScript, viii) Data Validation, ix) Confirmation Boxes.

Q.2 What do you mean by Java Script? Write its key features in detail.

Q.3 How to link Java Script files in Html?

Q.4 Explain external java script method to link file in HTML with example.

Q.5 Explain different types of variables used in java script.

Q.6 Write variables naming rules.

Q.7 Mentioned any 4 data types used in java script.

Q.8 Write difference between var, let and const in tabular form.

Q.9 Explain any four common string manipulation methods in JavaScript with suitable examples.

Q.10 Explain any four mathematical functions used in JavaScript with suitable examples.

Q.11 Explain how to call a function (calling function) in JavaScript with an example.

Q.12 What is return statement?

Q.13 Mention any five built-in objects in JavaScript with examples. How are they useful in JavaScript programming?

Q.14 Explain the role of Data Validation, Alert Messages, and Confirmation Boxes in JavaScript with suitable examples.



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UNIT-V

- Introduction to E-Business,
- Electronic Fund Transfer (EFT),
- Value chain Internet Business strategy,
- Functional Architecture,
- Implementation Strategies,
- Building Blocks of E-commerce,
- System design,
- Creating and managing content,
- Payment systems,
- Transaction Processing,
- E-commerce systems architecture.



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1. Introduction to E-business

E-business, short for electronic business, refers to the conduct of business activities using electronic means, primarily over the internet. It encompasses a wide range of activities, including buying and selling of goods and services, online transactions, electronic communication, and digital collaboration among businesses, customers, and other stakeholders. E-business has transformed the way organizations operate and interact with customers, offering numerous benefits such as increased efficiency, expanded market reach, and enhanced customer convenience.

Some key aspects of e-business:

- 1. Online Presence:** E-business starts with establishing an online presence, typically through a website or online storefront. This serves as the digital storefront where customers can browse products or services, place orders, and engage with the business.
- 2. E-Commerce:** One of the primary components of e-business is e-commerce, which involves buying and selling goods or services online. E-commerce platforms enable businesses to reach a global audience, streamline transactions, and offer various payment options to customers.
- 3. Digital Marketing:** E-business relies heavily on digital marketing strategies to attract and engage customers online. This includes techniques such as search engine optimization (SEO), social media marketing, email marketing, and content marketing to drive traffic to the website and promote products or services.
- 4. Customer Relationship Management (CRM):** E-business emphasizes building and maintaining strong relationships with customers through effective CRM strategies. This involves leveraging customer data, personalizing interactions, and providing excellent customer service to enhance satisfaction and loyalty.
- 5. Supply Chain Management:** E-business has revolutionized supply chain management by enabling real-time communication and collaboration among suppliers, manufacturers, distributors, and retailers. This leads to improved efficiency, inventory management, and responsiveness to customer demand.
- 6. Mobile Commerce:** With the proliferation of mobile devices, e-business has expanded to encompass mobile commerce or m-commerce. This involves conducting business transactions through mobile apps or optimized websites, catering to customers who prefer to shop and interact on their smartphones or tablets.
- 7. Data Analytics:** E-business generates vast amounts of data that can be analyzed to gain valuable insights into customer behavior, market trends, and business performance. Data

analytics tools help businesses make informed decisions, optimize processes, and personalize the customer experience.

8. Security and Privacy: E-business must prioritize security and privacy to protect sensitive information such as customer data, payment details, and intellectual property. Implementing robust security measures, encryption technologies, and compliance with regulations are essential to safeguarding against cyber threats and maintaining trust with customers.

9. Continuous Innovation: E-business is characterized by rapid technological advancements and evolving consumer preferences. Successful e-businesses continuously innovate and adapt to stay ahead of the competition, explore new markets, and leverage emerging technologies to enhance their offerings and operations.

2. Electronic Fund Transfer (EFT)

Electronic Fund Transfer, or EFT, is the electronic transfer of money between different bank accounts with no paper form like cheques or cash involved. It uses computer and electronic systems for transmitting funds in a secure and convenient manner.

2.1 Electronic Fund Transfer Process

An EFT transfer is usually very straightforward. There are two parties: the sender of funds and the receiver of funds. Once the sender initiates the transfer, the request channels through a series of digital networks from the Internet or a payment terminal to the sender's bank and then to the receiver's bank. Senders can be anyone from an employer to a business to an individual paying a vendor for a service such as electricity. Likewise, recipients can be entities like employees, goods suppliers, retailers, and utility companies. Most payments are cleared and completed within a couple of days.

2.2 Types of Electronic Fund Transfer (EFT)

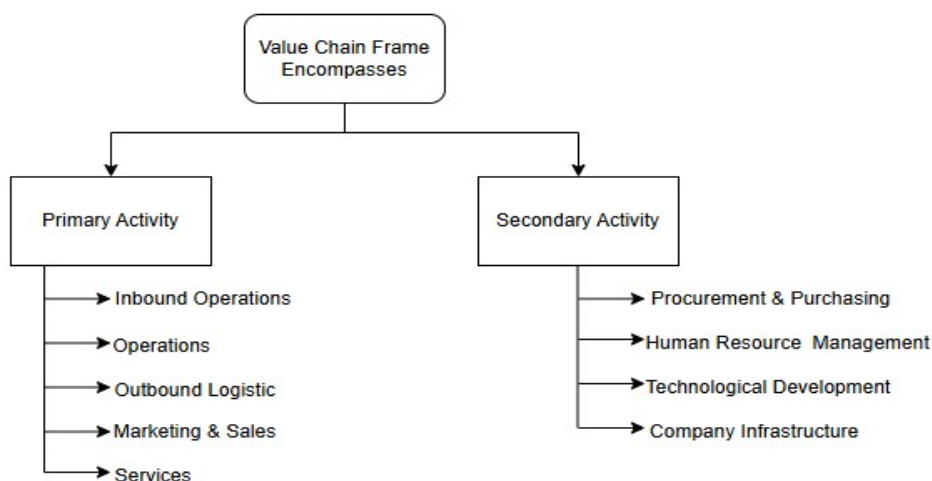
- **National Electronic Funds Transfer or NEFT:** An India-wide system of payments, enabling individuals and companies to shift funds from bank to bank on a batch basis.
- **RTGS (Real-Time Gross Settlement):** Used for high-value transactions, where the funds are settled in real-time and on a gross basis (not bundling).
- **IMPS (Immediate Payment Service):** Facilitates instant, 24x7 fund transfers through mobile or internet banking, appropriate for medium and large transfers.

- **UPI (Unified Payments Interface):** A mobile platform which facilitates instant transfers through a virtual payment address, UPI ID, or QR code.
- **Debit and Credit Card Transactions:** Involves electronic movement of funds while making purchases or cash withdrawals.
- **Wire Transfers:** Electronic money transfers made internationally by banks or money transfer services such as SWIFT.
- **Direct Deposit/Direct Debit:** Directly deposits wages or debits payments (such as bills) into/from the customer's bank account.

2.3 Benefits of EFT:

- **Speed and Convenience:** The funds are transferred quickly without the necessity of visiting a bank.
- **24/7 Availability:** Several EFT options such as UPI and IMPS are accessible 24/7.
- **Security:** The transactions are encrypted and controlled, allowing secure transfers of funds.
- **Less Paperwork:** Removes the necessity of cheques and paper documents.
- **Economical:** Low transaction costs in relation to traditional methods.
- **Error reduction:** Automation minimizes human error in manual transactions.

3. Value chain: A value chain is a concept describing the full chain of a business's activities in creating a product or service -- from initial receipt of materials through its delivery to market.



Value chain analysis occurs when a business identifies its primary and secondary activities and evaluates the efficiency of each point. This analysis can reveal linkages, dependencies and other patterns.

3.1 Primary Value Chain Activities (Main Business Functions)

These are the main steps involved in making, selling, and supporting a product or service:

- **Inbound Operations:** This includes receiving, handling, and storing materials that come from outside suppliers. These materials are called inputs and can include raw materials.
- **Operations:** These are the steps that turn the raw materials (inputs) into finished products or services (outputs) that can be sold to customers for a profit.
- **Outbound Logistics:** This involves delivering finished products to customers. It includes storing, packing, and transporting the goods.
- **Marketing and Sales:** These are efforts to promote the product or service, such as advertising and branding, to reach potential buyers and explain why they should purchase it.
- **Service:** After a sale, customer support and service help maintain a good relationship with customers by solving issues and offering help when needed.

When these activities are done efficiently, a business can reduce costs and gain an advantage over its competitors.

3.2 Secondary Value Chain Activities (Support Functions)

These activities support the primary functions and help the business run smoothly:

- **Procurement and Purchasing:** This means finding suppliers, maintaining good relationships with them, and buying the materials and resources needed to produce goods or services.
- **Human Resource Management (HRM):** This includes hiring, training, and keeping employees motivated and productive, and creating a positive workplace culture.
- **Technology Development:** This covers research, software and IT systems, cybersecurity, and other tech that helps the business operate and improve.
- **Company Infrastructure:** These are the essential internal services like legal, finance, administration, and quality control that help the business function effectively.

3.3 Benefits of Value Chain Analysis

- Make better decisions in key areas.
- Find and fix weaknesses or problems.
- Understand how different departments depend on each other.
- Improve efficiency and reduce costs.
- Compete more effectively by reducing expenses.
- Identify strengths and areas that need improvement.

4. Internet Business Strategy

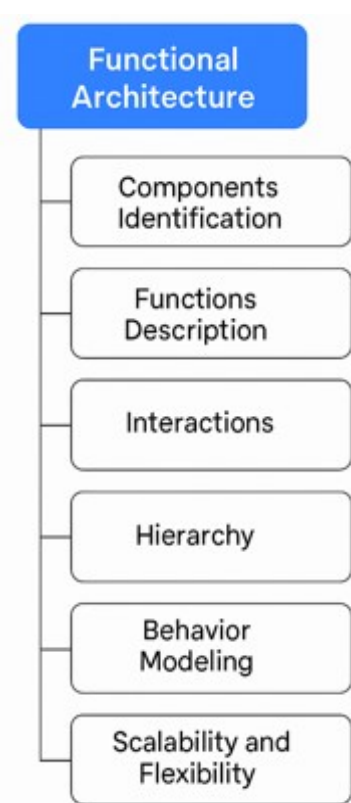
An internet business strategy is a company's plan to use the internet and digital tools to reach its goals. This includes areas like marketing, sales, customer support, product development, and operations.

4.1 Some good internet business strategy:

1. **Market Research:** Study your target customers – what they want, like, and how they behave online. This helps create products and services that meet their needs.
2. **Strong Online Presence:** Build a user-friendly website and stay active on social media. Make sure your website appears in search engine results (SEO).
3. **E-commerce Setup:** If you sell products or services online, build an e-commerce site. Include features like product listings, shopping carts, and secure payment options.
4. **Content Marketing:** Share useful and engaging content such as blogs, videos, and info graphics. Good content builds trust, brings in new visitors, and keeps customers coming back.
5. **Social Media Marketing:** Use platforms like Facebook, Instagram, or LinkedIn to promote your brand, connect with users, and run paid ads to reach more people.
6. **Email Marketing:** Send emails to share updates, promotions, and helpful tips. Tailor messages to different customer groups for better results.
7. **Data Analytics:** Track website traffic, customer behavior, and sales using data tools. Use the insights to improve your strategy and make better business decisions.
8. **Mobile Optimization:** Ensure your website and digital content work well on smartphones and tablets. A mobile-friendly site boosts customer satisfaction and sales.
9. **Online Customer Service:** Offer support through live chat, email, or social media. Quick and helpful responses improve customer satisfaction and build trust.
10. **Continuous Improvement:** Regularly update your strategy using customer feedback and performance data. Stay flexible and open to new technologies to stay ahead in the market.

5. Functional Architecture in E-Commerce

Functional architecture refers to how different parts of an e-commerce system work together to achieve specific tasks. It acts as a blueprint for building the system and includes the following:

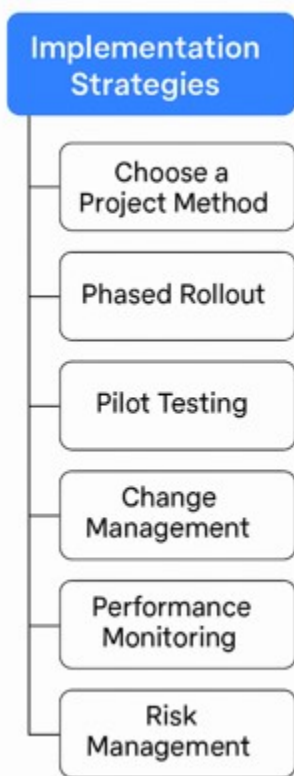


- 1. Components Identification:**
Identify all the important parts (modules) like user login, product search, shopping cart, payment, etc.
- 2. Functions Description:**
Define what each part does. For example, the shopping cart stores selected products, while the payment module handles money transactions.
- 3. Interactions:** Explain how different parts connect and communicate (e.g., checkout page sends data to payment gateway).
- 4. Hierarchy:** Break complex functions into smaller tasks for easier development and maintenance.
- 5. Behavior Modeling:** Use tools like flowcharts or diagrams to show how the system behaves step-by-step during user interaction.
- 6. Scalability and Flexibility:** Design the system so it can grow or be updated easily as business needs change.

6. Implementation Strategies for E-Commerce

These are the steps or plans used to successfully launch and manage an e-commerce platform:

- 1. Choose a Project Method:** Use methods like Agile (flexible and iterative) or Waterfall (step-by-step) to manage the project.
- 2. Phased Rollout:**
Start small (e.g., with a limited number of products or regions) and gradually expand.
- 3. Pilot Testing:**
Test the system with a small group of users to fix bugs and make improvements before full launch.



4. Change Management: Prepare employees and stakeholders for the new system through training, communication, and support.

5. Performance Monitoring: Track performance using KPIs (e.g., website speed, sales, user feedback) to ensure everything works smoothly.

6. Collect Feedback: Regularly ask users and staff for input to keep improving the system.

7. Risk Management: Identify potential risks (like security breaches or slow loading times) and create backup plans to deal with them.

7. Building Blocks of E-Commerce

These are the essential components needed to run an online business successfully:

- 1. Website or Online Platform:** This is your digital store where customers browse and buy your products or services.
- 2. Product Catalog:** A well-organized listing of your products or services, including pictures, descriptions, and prices.
- 3. Shopping Cart:** A virtual basket where users add products they want to buy before checking out.
- 4. Payment Gateway:** A secure system that handles online payments and keeps sensitive data (like card info) safe.
- 5. Shipping and Delivery:** Logistics that ensure orders are packed, shipped, and delivered to customers on time.
- 6. Security Measures:** Protect customer data with tools like SSL certificates, encryption, and secure logins.

7. **User Experience (UX) Design:** A clean, fast, and mobile-friendly website design that makes shopping easy and enjoyable.
8. **Customer Support:** Help centers or live chat that assist users with problems, returns, or questions.
9. **Marketing and Promotion:** Use SEO, social media, email, and discounts to attract and keep customers.
10. **Analytics and Reporting:** Track customer activity and sales data to make smart business decisions.

8. System Design

System design is the process of planning how different parts of a system—like architecture, components, modules, interfaces, and data—will work together to meet specific requirements.

It turns user needs into a detailed plan or blueprint, which helps developers build the system efficiently. The aim is to create a structure that is:

- Scalable (can grow with demand)
- Maintainable (easy to update and fix)
- High-performing (fast and reliable)

Note: A well-designed system is also fault-tolerant, meaning it can still function properly even if one or two parts fail.

8.1 Components of System Design





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- 1. Load balancers:** Most crucial component for scalability, availability, and performance measures for systems.
- 2. Key Value Stores:** It is a storage system similar to hash tables where key-value stores are distributed hash tables.
- 3. Blob Storage:** Blob stands for binary large objects, as the name suggests is storage for unstructured data such as YouTube, and Netflix.
- 4. Databases:** It is an organized collection of data so that they can be easily accessed and modified.
- 5. Rate Limiters:** These sets the maximum number of requests a service can fulfill.
- 6. Monitoring System:** These are basically software where system administrator monitor infrastructures such as bandwidth, CPU, routers, switches, etc.
- 7. Distributed System Messaging Queue:** Transaction medium between producers and consumers.
- 8. Distributed Unique ID generator:** In the case of large distributed systems, every moment multiple tasks are occurring so in order to distinguish it assign a tag corresponding to every event.
- 9. Distributed Search:** Over every website, crucial information that visitors will seek is put into the search bar.
- 10. Distributed Logging Services:** Tracing sequences of events from end to end.
- 11. Distributed Task Scheduler:** Computational resources such as CPU, memory, storage, etc.

9. Content Management (CM)

Content Management (CM) is the process of organizing, storing, sharing, and controlling access to information in various formats—like text, images, audio, video, and other media. It mostly refers to managing the entire digital content lifecycle, from creation to final deletion or archiving.

Stages of the Content Management Lifecycle

Seven stages of content management



Content management steps may vary based on the organization's needs, but the typical lifecycle includes:

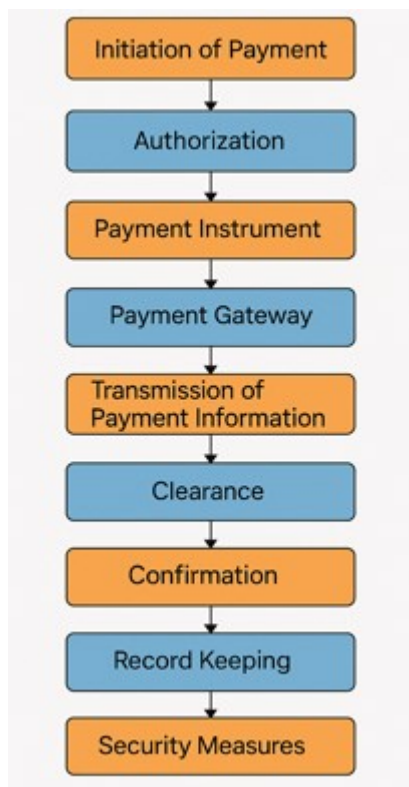
1. **Organization:** Plan how the content will be grouped and labeled. This includes creating categories, defining tags, and designing how content is classified.
2. **Creation:** Generate the content and assign it to the appropriate category or section based on its purpose or format.
3. **Storage:** Decide where and how the content will be stored (e.g., cloud, local server), focusing on security, accessibility, and delivery speed.
4. **Workflow:** Set rules and processes for how content moves through different roles or departments—ensuring consistency with company standards.
5. **Editing / Versioning:** Manage changes in content by tracking different versions and edits, especially when content is updated or improved over time.
6. **Publishing:** Make the content available to users—this could be publishing on a public website or sharing internally within a company intranet.
7. **Removal / Archiving:** Move old or unused content to an archive or delete it if it's no longer needed, ensuring the system stays organized and relevant.

10. Payment System

A payment system refers to a method that enables financial transactions between businesses and customers in a secure way. Nowadays, modern payment systems also provide additional features like fraud protection and analytics, allowing businesses to understand their customers' buying habits better. This helps companies create unique customer experiences that drive revenue and keep them competitive in the industry. By implementing the latest technology, businesses can benefit from efficient payment systems and save processing fees by paying in advance.

10.1 How Does the Payment System Work?

A payment system is a complex network of processes and entities that facilitate the transfer of funds between a payer (the person or entity making a payment) and a payee (the person or entity receiving the payment). The specifics can vary based on the type of payment system (e.g., traditional, electronic, mobile), but here is a general overview of how a payment system typically works:

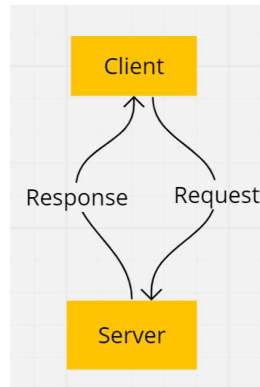


11. Types of E-Commerce Architecture

- Client-Server Architecture
- Two-Tier Architecture
- Three-Tier Architecture

11.1 Client-Server Architecture

In this architecture, the client (browser) sends the requests to the server, and the server processes the request if a request is valid then it responds with the requested data to the client. The client hosts the user interface (UI) while the server hosts the business logic and database.



Advantages of Client Server Architecture:

- There is a clear separation between the client side (user interface) and the server side (data and logic), making development more organized.
- It is easy to maintain and update the system.
- Data can be quickly and efficiently delivered to the client.

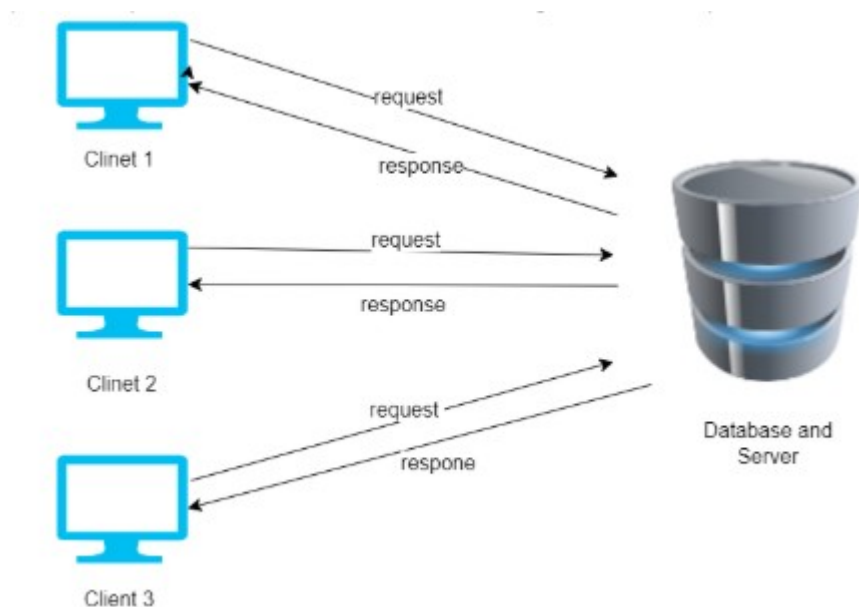
Disadvantages Client Server Architecture:

- If the server runs malicious code, client systems may be at risk of viruses or harmful scripts.
- Strong security measures are needed to prevent data tampering or interception during transmission.
- If the server goes down, clients cannot connect or access any data until it is restored.

11.2 Two-Tier Architecture

The two-tier architecture has consisted of mainly two components:

1. **Client layer:** It consists of the web browser, mobile application, or the other UI that user interacts with. This front-end client makes requests to the server.
2. **Server layer:** It handles both the application logic and data storage/management. This single back-end server acts as both the application server and the database server.



Advantages Two- Tier Architecture:

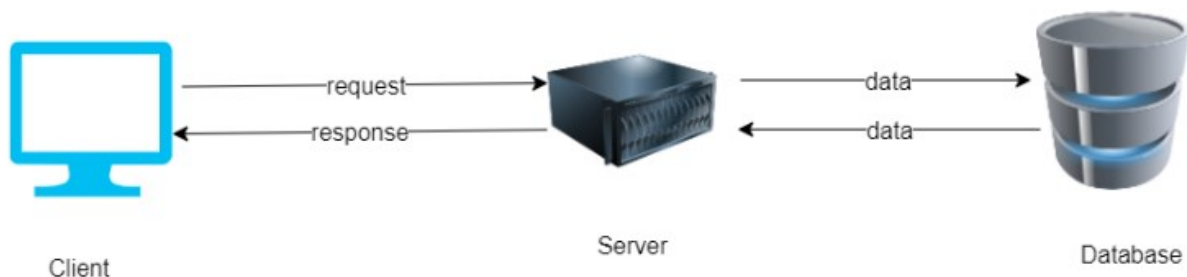
- It is simple to develop and deploy
- The client only communicates with one backend system
- All data logic and validation is handled on the server

Disadvantages Two- Tier Architecture:

- Server handles too many tasks, causing performance issues.
- Limited scalability due to tightly connected layers.
- Low flexibility as UI and logic are coupled.

11.3 Three-Tier Architecture

Three-tier architecture is widely used in building e-commerce websites. It solves the limitations of two-tier architecture by clearly separating the system into three layers:



1. Client Tier (Presentation Layer): This is the front-end that users interact with—like a web browser or mobile app. It sends requests to the server and displays the results.

2. Middle Tier (Application Layer): This is where the business logic is handled. It processes user requests, interacts with the database, performs calculations, and returns responses.

3. Data Tier (Database Layer): This back-end layer stores and manages data using databases like Oracle, MySQL, or MongoDB. It responds to queries from the application layer.

Advantages Three-Tier Architecture:

- Clear separation of functions makes it easy to manage and maintain.
- Each layer can be scaled separately to handle more users or data.
- High scalability with multiple app and database servers.
- Supports redundancy and failover, improving reliability.

Disadvantages Three-Tier Architecture:

- More complex setup; managing three layers can be harder, especially for small apps.
- Extra layers can slow down performance due to more data transfer.
- Scaling all layers efficiently may require more time and resources.
- More communication between layers can lead to delays or latency.



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UNIT RELATED QUESTIONS

Q.1 Define: i) E-Commerce, ii) Digital Marketing, iii) Supply Chain Management, iv) Customer Relationship Management (CRM), v) Content Management (CM),

Q.2 What is ETF (Electronic Fund Transfer)? Describe its types in detail.

Q.3 What is value chain? Describe primary value chain activities & Secondary value chain activities in detail.

Q.4 Write a short note on Value chain including its benefits.

Q.5 Write some good internet business strategies.

Q.6 What do you mean by functional architecture for e- commerce.

Q.7 Write implementation strategies for e- commerce in detail.

Q.8 Describe building block of e- commerce in detail.

Q.9 Explain system design including its aim and components in detail.

Q.10 Describe stages of the Content Management Lifecycle in detail.

Q.11 What do you mean by payment system? How does it work?

Q.12 Explain client server architecture including its advantages, disadvantages& diagram in detail.

Q.13 Explain two- tier architecture including its advantages, disadvantages& diagram in detail.

Q.14 Explain three- tier architecture including its advantages, disadvantages& diagram in detail.

Q.15 Mention types of E-Commerce architecture and explain any one architecture including its advantages, disadvantages& diagram in detail.

Assignments

Note: Submit all assignment in office file. (Attach front page)

Last Date of Submission 2-06-2025

UNIT-1 (Attempt any 5)

Q.1 Define: i) Internet, ii) Intranet, iii) URL, iv) DNS, v) HTTP, vi) HTTPS, vii) FTP, viii) SMTP, ix) PPP, x) TCP/IP.

Q.2 Write the difference between internet and intranet.

Q.3 What is internet? Explain its working with diagram & also mention the advantages, disadvantages & applications of it.

Q.4 Write a short note on remote logging & local logging in TELNET with diagram.

Q.5 Write a short note on Internet Protocol & its working with diagram.

UNIT-2 (Attempt any 5)

Q.1 What is the difference between an HTML element and an HTML attribute & HTML Tags?

Q.2 Explain the basic structure of HTML with purpose of each section.

Q.3 Write short notes on i) Radio Button Control, ii) Password field Control, iii) Check-box Control, iv) Submit Button, v) Textarea, vi) Input element, vii) Label Element.

Q.4 Define following HTML tags. For each, write the **tag name**, its **purpose (definition)**, **syntax**, and **one example**: , , <i>, , <mark>, <u>, <strike>, <sup>, , <ins>, <big>, <small>, , , <dl>, <dd>, <dt>, , <h1> to <h6>, <p>, <a>, , <div>, , , , , <table>, <tr>, <td>, <form>, and <input>,
.

Q.5 Write HTML code to generate a table with 3 rows and 3 columns, including custom data in each cell.

Q.6 Create an HTML document that displays an ordered list with a minimum of three list items.

Q.7 Create an HTML document that displays an un-ordered list with a minimum of three list items.

UNIT-3 (Attempt any 5)

Q.1 Write the definition, syntax and example of following properties: background- color, background- image, background-repeat, background – attachment, background- position.

Q.2 What are the three different ways to apply CSS to an HTML document, and how does each method work?

Q.3 Write the differences between (i) Block elements, (ii) Inline elements, and (iii) Inline-block elements.

Q.4 Write any 7 CSS font properties with definition, syntax and example.

Q.5 Write any 7 CSS text formatting properties with definition, syntax and example.

Q.6 What is a CSS list, how many types of lists are there in HTML, and what are the main CSS list properties along with their definitions?

Q.7 Define: i) Margin, ii) Padding, iii) Any 3 Color Specification, iv) Id selector, v) Class selector, vi) Unordered List in CSS, vii) Ordered List in CSS

UNIT-4 (Attempt any 5)

Q.1 Define: i) Java Script, ii) String Concatenation, iii) Imperative Language, iv) Declarative Language, v) Return Statement, vi) Function Parameter, vii) Variable in JavaScript, viii) Data Validation, ix) Confirmation Boxes.

Q.2 What do you mean by Java Script? Write its key features in detail.

Q.3 How to link Java Script files in Html? Explain both ways in detail.

Q.4 Explain all string manipulation methods in JavaScript with suitable examples.

Q.5 Explain the role of Data Validation, Alert Messages, and Confirmation Boxes in JavaScript with suitable examples.

Q.6 Mention any two difference between var, let and const in tabular form.

UNIT-5 (Attempt any 5)

Q.1 Define: i) E-Commerce, ii) Digital Marketing, iii) Supply Chain Management, iv) Customer Relationship Management (CRM), v) Content Management (CM).

Q.2 Mention all types of e-commerce architecture. Explain any one of them in detail including its advantages, disadvantages & diagram in detail.

Q.3 What is ETF (Electronic Fund Transfer)? Describe its types in detail.

Q.4 Write a short note on Value chain including its benefits.

Q.5 Write some good internet business strategies. (At least 8)

Q.6 Explain system design including its aim and components in detail.

Q.7 Describe stages of the Content Management Lifecycle in detail.